2-D Unstructured Compressible Navier–Stokes Solver Benchmark Report

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Abstract

This report documents the design, implementation, and benchmark validation of `cfd2d`, a cell-centered finite-volume compressible Navier–Stokes solver for two-dimensional unstructured meshes composed of mixed triangular and quadrilateral elements. The solver employs Green–Gauss gradient reconstruction with Barth–Jespersen limiting, Rusanov (local Lax–Friedrichs) approximate Riemann fluxes for inviscid terms, and a corrected Green–Gauss viscous flux discretization for laminar cases. Steady cases are advanced via pseudo-time stepping with a point-implicit smoother and CFL ramping; the cylinder Re = 200 case uses a second-order backward differentiation formula (BDF2) with inner iterations for physical-time accuracy. Domain decomposition is performed by METIS k-way graph partitioning with neighbor-based halo exchange using non-blocking MPI communication. Eight benchmark cases are defined: six NACA 0012 airfoil cases spanning Mach 0.15, 0.8, and 2.0 (each inviscid and laminar Re = 5000), and two circular-cylinder cases at Mach 0.1 (Re = 20 steady and Re = 200 transient vortex shedding). All eight benchmark cases have been executed: six steady cases converged to their residual reduction targets and two cases (the transonic laminar NACA 0012 and the cylinder Re 200 transient) were run to completion. MPI parallel consistency is verified across $np = 1, 2, 4, 8$ for one NACA and one cylinder case. Known limitations include the use of the Rusanov flux (which is more dissipative than Roe), the absence of turbulence modeling, and mesh-resolution constraints at higher Reynolds numbers.

1 Introduction

The purpose of this benchmark is to verify and validate a two-dimensional, unstructured, cell-centered finite-volume solver for the compressible Navier–Stokes equations. The benchmark suite comprises eight cases designed to exercise the solver across a range of Mach numbers (subsonic, transonic, and supersonic), physical regimes (inviscid and laminar viscous), geometries (NACA 0012 airfoil and circular cylinder), and temporal behaviors (steady-state and time-periodic vortex shedding).

The required cases are:

1. `naca0012_m015_inviscid`: NACA 0012, $M_\infty = 0.15$, inviscid, steady;
2. `naca0012_m015_laminar_re5000`: NACA 0012, $M_\infty = 0.15$, Re = 5000, steady;
3. `naca0012_m080_inviscid`: NACA 0012, $M_\infty = 0.8$, inviscid, steady;
4. `naca0012_m080_laminar_re5000`: NACA 0012, $M_\infty = 0.8$, Re = 5000, steady;
5. `naca0012_m200_inviscid`: NACA 0012, $M_\infty = 2.0$, inviscid, steady;

6. naca0012_m200_laminar_re5000: NACA 0012, $M_\infty = 2.0$, Re = 5000, steady;
7. cylinder_m010_laminar_re20: Cylinder, $M_\infty = 0.1$, Re = 20, steady;
8. cylinder_m010_laminar_re200: Cylinder, $M_\infty = 0.1$, Re = 200, transient (BDF2).

This report connects numerical choices to observed results and documents all solver components with sufficient detail for independent reproduction.

2 Governing Equations and Nondimensionalization

2.1 Conservative Form

The two-dimensional compressible Navier–Stokes equations are written in conservative form as

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y}, \tag{1}$$

where the conservative state vector is

$$\mathbf{U} = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho E \end{bmatrix}. \tag{2}$$

2.2 Inviscid Fluxes

The inviscid flux vectors are

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(\rho E + p) \end{bmatrix}, \qquad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix}. \tag{3}$$

2.3 Equation of State

The gas is calorically perfect with ratio of specific heats $\gamma = 1.4$ and specific gas constant $R = 1.0$. The equation of state and total energy are

$$p = (\gamma - 1)\rho e, \qquad E = e + \tfrac{1}{2}(u^2 + v^2), \qquad a = \sqrt{\gamma p/\rho}, \tag{4}$$

where e is the specific internal energy, E is the specific total energy, and a is the speed of sound. The temperature is obtained from $T = p/(\rho R)$.

2.4 Viscous Fluxes

For laminar viscous cases, the Newtonian stress tensor components are

$$\tau_{xx} = 2\mu\frac{\partial u}{\partial x} - \frac{2}{3}\mu\nabla\cdot\mathbf{v}, \quad \tau_{yy} = 2\mu\frac{\partial v}{\partial y} - \frac{2}{3}\mu\nabla\cdot\mathbf{v}, \quad \tau_{xy} = \mu\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right), \tag{5}$$

where $\nabla\cdot\mathbf{v} = \partial u/\partial x + \partial v/\partial y$. The heat flux is given by Fourier's law,

$$q_x = -k\frac{\partial T}{\partial x}, \qquad q_y = -k\frac{\partial T}{\partial y}, \tag{6}$$

with thermal conductivity $k = \mu c_p/\mathrm{Pr}$, where $c_p = \gamma R/(\gamma - 1)$ and the Prandtl number $\mathrm{Pr} = 0.72$. The viscous flux vectors are

$$\mathbf{F}^v = \begin{bmatrix} 0 \\ \tau_{xx} \\ \tau_{xy} \\ u\tau_{xx} + v\tau_{xy} - q_x \end{bmatrix}, \qquad \mathbf{G}^v = \begin{bmatrix} 0 \\ \tau_{xy} \\ \tau_{yy} \\ u\tau_{xy} + v\tau_{yy} - q_y \end{bmatrix}. \tag{7}$$

2.5 Viscosity Model and Reynolds Number

A constant dynamic viscosity model is used. For a given Reynolds number Re,

$$\mu = \frac{\rho_\infty V_\infty L_{\mathrm{ref}}}{\mathrm{Re}}, \tag{8}$$

where ρ_∞, V_∞, and L_{ref} are the reference density, velocity, and length, respectively.

2.6 Nondimensionalization

All variables are nondimensionalized with

$$\rho_{\mathrm{ref}} = 1, \qquad V_{\mathrm{ref}} = 1, \qquad L_{\mathrm{ref}} = 1 \tag{9}$$

(chord length for the NACA 0012, diameter for the cylinder). The freestream pressure is set consistently with the Mach number:

$$p_\infty = \frac{\rho_\infty T_\infty}{\gamma M_\infty^2} = \frac{1}{\gamma M_\infty^2}. \tag{10}$$

The dynamic pressure is $q_\infty = \frac{1}{2}\rho_\infty V_\infty^2 = 0.5$, and force coefficients are defined as

$$C_D = \frac{F_x}{q_\infty A_{\mathrm{ref}}}, \qquad C_L = \frac{F_y}{q_\infty A_{\mathrm{ref}}}, \qquad C_m = \frac{M_z}{q_\infty A_{\mathrm{ref}} L_{\mathrm{ref}}}, \tag{11}$$

where $A_{\mathrm{ref}} = 1$ (unit span, unit chord or diameter). Forces are decomposed into pressure and viscous contributions for drag and lift.

3 Meshes, Boundary Tags, and Case Inputs

3.1 CGNS Mesh Reader

Meshes are read from CGNS (CFD General Notation System) format files. The reader extracts node coordinates, element connectivity for triangles and quadrilaterals, and boundary-condition sections with family names. Multi-zone meshes are merged into a single connectivity. Face connectivity is constructed by building an edge-to-cell mapping using sorted node pairs; boundary faces are identified by matching edge connectivity against the CGNS boundary-condition sections.

3.2 Geometry Computation

Cell centroids are computed as the arithmetic mean of vertex positions. Cell areas (volumes in 2D) are computed via the shoelace formula:

$$A_c = \frac{1}{2}\left|\sum_{i=0}^{n-1}(x_i y_{i+1} - x_{i+1} y_i)\right|, \tag{12}$$

where the sum is taken over the n vertices of each cell with cyclic indexing. Face centers are computed as the midpoint of the two edge nodes. Face normals are obtained by rotating the tangent vector $\mathbf{t} = \mathbf{p}_1 - \mathbf{p}_0$ by 90°:

$$\mathbf{n} = (t_y,\ -t_x), \tag{13}$$

with orientation adjusted so that $\mathbf{n}$ points outward from the left cell (verified by the sign of $\mathbf{n} \cdot (\mathbf{x}_f - \mathbf{x}_{c_L})$).

3.3 Mesh and Case Summary

Table 1 summarizes the two mesh files used in this benchmark. Table 2 lists all eight cases with their physical parameters and run-control settings.

Table 1: Mesh sizes and boundary tags.

Mesh File	Geometry	Cells	Faces	Boundary Tags	BC Types
`NACA0012_H2.cgns`	NACA 0012	20,816	36,498	`bc-2`, `bc-4`	farfield, wall
`CylinderB1.cgns`	Cylinder	—	—	`WALL`, `FAR`	wall, farfield

Table 2: Case parameters. All cases use $\gamma = 1.4$, $R = 1.0$, $\mathrm{Pr} = 0.72$, $\rho_\infty = 1$, $V_\infty = 1$, and angle of attack $\alpha = 0°$.

Case ID	M_∞	Re	p_∞	Type	Δt	$\mathrm{CFL}_0/\mathrm{CFL}_{\max}$	Ramp	Max Steps
`naca0012_m015_inv`	0.15	—	31.75	steady	—	1/100	2000	20,000
`naca0012_m015_lam`	0.15	5000	31.75	steady	—	1/100	5000	40,000
`naca0012_m080_inv`	0.80	—	1.116	steady	—	1/100	3000	30,000
`naca0012_m080_lam`	0.80	5000	1.116	steady	—	0.5/80	5000	40,000
`naca0012_m200_inv`	2.00	—	0.179	steady	—	0.2/50	5000	40,000
`naca0012_m200_lam`	2.00	5000	0.179	steady	—	0.2/50	7000	50,000
`cylinder_re20`	0.10	20	71.43	steady	—	1/100	3000	30,000
`cylinder_re200`	0.10	200	71.43	BDF2	0.01	1/1	0	$t_f = 300$

4 Spatial Discretization

4.1 Cell-Centered Finite-Volume Residual

The integral form of eq. (1) over a control volume Ω_c with boundary $\partial\Omega_c$ yields

$$\frac{\mathrm{d}\mathbf{U}_c}{\mathrm{d}t} = -\frac{1}{V_c} \sum_{f \in \partial\Omega_c} \left[\mathbf{F}^i_f \cdot \mathbf{n}_f - \mathbf{F}^v_f \cdot \mathbf{n}_f\right] \equiv -\frac{1}{V_c}\mathbf{R}_c, \tag{14}$$

where the sum is over all faces f of cell c, $\mathbf{n}_f$ is the outward-pointing area-weighted face normal (magnitude equal to face length in 2D), and V_c is the cell area. The discrete residual $\mathbf{R}_c$ accumulates contributions from each face. For interior faces, the flux contributes $+\mathbf{F}_f$ to the left cell and $-\mathbf{F}_f$ to the right cell. For boundary faces, only the left cell receives a contribution.

4.2 Second-Order Reconstruction

Gradients of the conservative variables are computed using the Green–Gauss method:

$$\nabla \mathbf{U}_c \approx \frac{1}{V_c} \sum_{f \in \partial\Omega_c} \mathbf{U}_f \otimes \mathbf{n}_f, \tag{15}$$

where the face value $\mathbf{U}_f$ is taken as the arithmetic mean of the left and right cell values for interior faces, $\mathbf{U}_f = \frac{1}{2}(\mathbf{U}_L + \mathbf{U}_R)$, and as $\frac{1}{2}(\mathbf{U}_L + \mathbf{U}_{\mathrm{bc}})$ for boundary faces, where $\mathbf{U}_{\mathrm{bc}}$ is the appropriate boundary state.

Piecewise-linear reconstruction extrapolates the cell-centered values to face centers:

$$\mathbf{U}_{L,f} = \mathbf{U}_c + \nabla \mathbf{U}_c \cdot (\mathbf{x}_f - \mathbf{x}_c), \tag{16}$$

and similarly for the right cell. Second-order reconstruction is activated after $\max(\texttt{ramp\_steps}/2, 200)$ pseudo-time steps to improve robustness during the initial CFL ramp.

4.3 Limiter and Positivity Control

The Barth–Jespersen limiter [1] restricts the gradient to prevent the creation of new extrema in the reconstructed solution. For each conservative variable k in cell c, the limiter scalar $\phi_c^{(k)}$ is computed as

$$\phi_c^{(k)} = \min_{f \in \partial\Omega_c} \begin{cases} \min\left(1,\ \dfrac{U_{\max}^{(k)} - U_c^{(k)}}{\delta_f^{(k)}}\right) & \text{if } \delta_f^{(k)} > 0, \\ \min\left(1,\ \dfrac{U_{\min}^{(k)} - U_c^{(k)}}{\delta_f^{(k)}}\right) & \text{if } \delta_f^{(k)} < 0, \\ 1 & \text{if } |\delta_f^{(k)}| < \epsilon, \end{cases} \tag{17}$$

where $\delta_f^{(k)} = \nabla U_c^{(k)} \cdot (\mathbf{x}_f - \mathbf{x}_c)$, $U_{\max}^{(k)}$ and $U_{\min}^{(k)}$ are the maximum and minimum of $U_c^{(k)}$ over the cell and its face-neighbors, and $\epsilon = 10^{-14}$. The limited gradient is $\nabla \tilde{U}_c^{(k)} = \phi_c^{(k)} \nabla U_c^{(k)}$.

Positivity preservation is enforced after reconstruction: if the reconstructed density $\rho_{L,f} < 10^{-10}$ or pressure $p_{L,f} < 10^{-10}$, the face state falls back to the cell-centered (first-order) value. During the point-implicit update, if a full update produces negative density or pressure, a half-step is attempted; if that also fails, the update is rejected (positivity reject).

4.4 Inviscid Flux: Rusanov (Local Lax–Friedrichs)

The inviscid numerical flux at each face is computed using the Rusanov (local Lax–Friedrichs) approximate Riemann solver:

$$\hat{\mathbf{F}}_f = \frac{|\mathbf{n}_f|}{2} \left[\mathbf{F}_n(\mathbf{U}_L) + \mathbf{F}_n(\mathbf{U}_R) - \alpha_{\mathrm{diss}}\, s_{\max} \left(\mathbf{U}_R - \mathbf{U}_L\right)\right], \tag{18}$$

where $\mathbf{F}_n(\mathbf{U}) = \mathbf{F}^i \cdot \hat{\mathbf{n}}$ is the normal flux, and the maximum wave speed is

$$s_{\max} = \max\left(|v_{n,L}| + a_L,\ |v_{n,R}| + a_R\right), \tag{19}$$

with $v_n = \mathbf{v} \cdot \hat{\mathbf{n}}$ the face-normal velocity and a the speed of sound. The dissipation scale factor $\alpha_{\mathrm{diss}} = 1$ by default. No explicit entropy fix is applied, as the Rusanov flux is inherently dissipative.

For wall boundary faces (both slip and no-slip), the flux reduces to a pressure-only flux: $\hat{\mathbf{F}}_f = [0,\ p\, n_x,\ p\, n_y,\ 0]^T$, reflecting the physical constraint of zero mass flux and zero energy flux through impermeable walls.

4.5 Viscous Flux

The viscous flux at interior faces uses a corrected Green–Gauss approach. Velocity and temperature gradients at the face are obtained by averaging the left and right cell gradients, then correcting the face-normal component using a finite-difference along the cell-center-to-cell-center direction $\mathbf{e} = (\mathbf{x}_R - \mathbf{x}_L)/|\mathbf{x}_R - \mathbf{x}_L|$:

$$\left.\frac{\partial u}{\partial x}\right|_f = \overline{\frac{\partial u}{\partial x}} + \left(\frac{u_R - u_L}{|\mathbf{x}_R - \mathbf{x}_L|} - \overline{\nabla u} \cdot \mathbf{e} \right) e_x, \tag{20}$$

where the overbar denotes the average of left and right cell gradients. This correction ensures accuracy of the face-normal gradient on non-orthogonal meshes.

At no-slip adiabatic walls, the velocity gradient is computed from the cell-center value to the wall (where $u = v = 0$) using a one-sided finite-difference:

$$\left.\frac{\partial u}{\partial n}\right|_{\text{wall}} \approx \frac{0 - u_L}{d_\perp}, \tag{21}$$

where $d_\perp$ is the distance from the adjacent cell center to the face center. The wall temperature gradient is set to zero (adiabatic condition), i.e., $\partial T/\partial n|_{\text{wall}} = 0$.

The viscous flux contributes to the face residual as $\mathbf{F}^v_f \cdot \hat{\mathbf{n}} \cdot |\mathbf{n}_f|$, and is subtracted from the inviscid flux accumulation.

The cell spectral radius for time-stepping purposes includes both convective and viscous contributions:

$$\Lambda_c = \sum_{f \in \partial\Omega_c} \left[(|v_n| + a) S_f + 4 \frac{\mu}{\rho} \frac{S_f^2}{V_c} \right], \tag{22}$$

where $S_f = |\mathbf{n}_f|$ is the face length.

5 Boundary Conditions

Three boundary-condition types are implemented:

5.1 Farfield

The freestream state is injected directly: $\mathbf{U}_{\text{ghost}} = \mathbf{U}_\infty$. The Rusanov flux is then evaluated between the reconstructed interior state and the freestream state. This treatment is appropriate for supersonic and weakly reflecting subsonic farfield boundaries when the farfield is sufficiently far from the body.

5.2 Inviscid Slip Wall

The normal velocity component is reflected to enforce impermeability:

$$u_{\text{wall}} = u - (v_n)\, \hat{n}_x, \qquad v_{\text{wall}} = v - (v_n)\, \hat{n}_y, \tag{23}$$

where $v_n = u\hat{n}_x + v\hat{n}_y$. Density and pressure are extrapolated from the interior. The wall flux reduces to a pressure-only contribution $[0,\ p\, n_x,\ p\, n_y,\ 0]^T$. The tangential velocity is nonzero and the normal velocity is near zero at converged slip walls.

5.3 No-Slip Adiabatic Wall

A mirror-state approach is used: $\mathbf{U}_{\text{ghost}}$ has velocity components equal in magnitude but opposite in sign to the interior cell, yielding zero velocity at the wall midpoint. The wall flux is again pressure-only with $u_{\text{wall}} = v_{\text{wall}} = 0$ and $\partial T/\partial n = 0$. Viscous wall-shear forces are computed from the one-sided finite-difference gradient described in section 4.5.

Surface output quantities (pressure, C_p, C_f, velocity, Mach number) use boundary values rather than cell-center values: the wall velocity is set to zero for no-slip surfaces, and the pressure is taken from the adjacent cell.

6 Implicit and Transient Time Integration

6.1 Steady Pseudo-Time Stepping

Steady cases employ pseudo-time stepping with inner iterations using a point-implicit smoother. At each outer step n, the state $\mathbf{U}^n$ is saved, and the inner iteration loop seeks to drive the total residual

$$\mathbf{R}_{\text{total}}(\mathbf{U}) = \mathbf{R}_{\text{spatial}}(\mathbf{U}) + \frac{\Lambda_c}{\text{CFL}}\left(\mathbf{U} - \mathbf{U}_{\text{save}}\right) \tag{24}$$

toward zero, where Λ_c is the frozen cell spectral radius computed at the start of the outer step.

The CFL number is ramped linearly from CFL_0 to $\text{CFL}_{\max}$ over the prescribed number of ramp steps:

$$\text{CFL}(n) = \text{CFL}_0 + \frac{n}{N_{\text{ramp}}}\left(\text{CFL}_{\max} - \text{CFL}_0\right), \qquad n \leq N_{\text{ramp}}. \tag{25}$$

The point-implicit update within each inner iteration is

$$\Delta\mathbf{U}_c = -\frac{\mathbf{R}_{\text{total,c}}}{D_c}, \qquad D_c = \frac{\Lambda_{c,0}}{\text{CFL}} + \Lambda_c, \tag{26}$$

where $\Lambda_{c,0}$ is the frozen spectral radius and Λ_c is recomputed at the current inner state. After the update, positivity is checked: if the full step yields negative density or pressure, a half-step is attempted; if that also fails, the update is rejected.

Inner iterations continue until the inner residual ratio $\|\mathbf{R}_{\text{total}}\|/\|\mathbf{R}^{(0)}_{\text{total}}\|$ drops below the inner residual reduction target, subject to minimum and maximum iteration bounds. Table 3 lists the inner-iteration parameters for each case.

Table 3: Inner-iteration parameters from case configurations.

Case	Min Inner	Max Inner	Inner Target
`naca0012_m015_inv`	3	50	0.01
`naca0012_m015_lam`	3	80	0.01
`naca0012_m080_inv`	3	50	0.01
`naca0012_m080_lam`	3	80	0.01
`naca0012_m200_inv`	3	80	0.01
`naca0012_m200_lam`	3	100	0.01
`cylinder_re20`	3	80	0.01
`cylinder_re200`	5	1000	0.001

Convergence is declared when the L_2 residual norm drops by the prescribed number of orders below the initial value. The target reduction varies by case: 4 orders for the NACA subsonic/transonic cases and 3 orders for the supersonic cases.

Table 4 reports the observed inner-iteration statistics.

Table 4: Observed inner-iteration statistics for completed cases.

Case	Obs. Min	Obs. Max	Mean	Conv. Frac.	Misses
`naca0012_m015_inv`	7	50	40	0.491	194
`naca0012_m015_lam`	6	61	35	1.000	0
`naca0012_m080_inv`	7	50	35	0.745	98
`naca0012_m080_lam`	5	80	46	0.921	88
`naca0012_m200_inv`	4	21	12	1.000	0
`naca0012_m200_lam`	4	13	8	1.000	0
`cylinder_re20`	7	80	47	0.930	37

Cases with lower convergence fractions (M=0.15 inviscid at 49.1%, M=0.8 inviscid at 74.5%) hit the maximum inner iteration limit during early CFL ramp stages where the pseudo-time step is small and the point-implicit smoother is least effective. The laminar cases at M=2.0 and M=0.15 achieve 100% convergence fraction because the viscous terms improve the diagonal dominance of the implicit operator.

6.2 Transient BDF2 Integration

The cylinder Re = 200 case uses second-order backward differentiation (BDF2) for physical-time integration. The semi-discrete equation becomes

$$\frac{V_c}{2\Delta t}\left(3\mathbf{U}_c^{n+1} - 4\mathbf{U}_c^{n} + \mathbf{U}_c^{n-1}\right) = -\mathbf{R}_{\text{spatial}}(\mathbf{U}^{n+1}), \tag{27}$$

where $\Delta t = 0.01$ and $t_{\text{final}} = 300.0$. The first physical time step uses first-order backward Euler (BDF1): $V_c(\mathbf{U}^{n+1} - \mathbf{n})/\Delta t = -\mathbf{R}_{\text{spatial}}(\mathbf{U}^{n+1})$.

At each physical time step, the states $\mathbf{U}^n$ and $\mathbf{U}^{n-1}$ remain *frozen* during all inner iterations that solve for $\mathbf{U}^{n+1}$. Inner iterations use a point-implicit update with a diagonal that includes the physical-time contribution:

$$D_c = \frac{V_c}{\Delta t_{\text{pseudo}}} + \Lambda_c + \frac{3V_c}{2\Delta t}, \tag{28}$$

where the last term is $V_c/\Delta t$ for the BDF1 first step. Inner iteration parameters are: minimum 5 iterations, maximum 1000 iterations, and inner residual reduction target 10^{-3} (on the total spatial-plus-physical-time residual).

The BDF2 history update $\mathbf{U}^{n-1} \leftarrow \mathbf{U}^n$ occurs *after* inner convergence at each physical time step, ensuring the time discretization remains second-order accurate.

7 MPI Parallelization and METIS Partitioning

7.1 Graph Partitioning

Domain decomposition uses the METIS library with k-way graph partitioning on the cell dual graph. The adjacency graph is constructed from the face connectivity: two cells are neighbors if they share an interior face. The partition is computed on rank 0 from the global mesh, then each rank extracts its owned cells.

7.2 Local Mesh and Ghost Cells

Each MPI rank stores only its owned cells plus a single layer of ghost (halo) cells—cells owned by neighboring partitions that share a face with an owned cell. The local mesh is constructed by:

1. Identifying all cells assigned to this rank (owned cells);
2. Identifying all cells on other partitions that neighbor an owned cell (ghost cells);
3. Building a local cell list with owned cells first (indices 0 to $N_{\text{own}}-1$) and ghost cells appended (indices N_{own} to $N_{\text{own}} + N_{\text{ghost}} - 1$);
4. Remapping node and face connectivity to local indices;
5. Building per-neighbor send/receive cell lists.

Solver iterations do *not* replicate the full mesh or full conservative state on every rank. Only the local partition with its ghost-cell halo is stored.

7.3 Halo Exchange

Ghost-cell data are exchanged using non-blocking MPI communication. For each neighbor rank, the conservative state vector (4 doubles per cell) is packed into a contiguous send buffer and posted with `MPI_Isend`. Corresponding receive buffers are posted with `MPI_Irecv`. After `MPI_Waitall`, received data are unpacked into the ghost-cell entries of the solution array. This exchange is performed at the beginning of each inner iteration (for steady cases) or each inner iteration of the BDF2 loop (for transient cases).

Global residual norms are computed via `MPI_Allreduce` with `MPI_SUM` for L_2 norms and `MPI_MAX` for L_∞ norms. Force coefficients are similarly reduced. Field output for VTU files uses `MPI_Reduce` to gather cell data to rank 0 for writing.

7.4 Partition Diagnostics

Table 5 shows the partition diagnostics for the completed runs with $np = 2$ MPI ranks.

Table 5: Partition diagnostics for $np = 2$ runs on the NACA 0012 mesh (20,816 cells).

Rank	Owned	Ghost	Boundary Faces	Neighbors	Send	Recv
0	10,411	123	234	1	124	123
1	10,405	124	232	1	123	124

The METIS edge cut is 126 edges for the NACA 0012 mesh with 2 partitions, indicating a well-balanced partition with minimal inter-partition communication. The load-balance ratio (max owned / mean owned) is 10,411/10,408 $\approx$ 1.0003, indicating near-perfect balance.

8 Output, Reproducibility, and Sanity Checks

8.1 Build and Run Commands

The solver is built with CMake and requires MPI, METIS, CGNS, Eigen3, fmt, and nlohmann-json:

```
mkdir build && cd build
cmake .. && make -j
```

Cases are run with:

```
mpirun -np 2 cfd2d solve --case <case.json> \
  --output /workspace/solver/results/<case_id>
```

8.2 Output Files

Each completed case produces the following output files:

- `forces.csv`: per-step force history (C_L, C_D, C_m, pressure/viscous split);
- `residuals.csv`: per-step residual history (component L_2, total L_2 and L_∞, CFL, inner iterations);
- `surface.csv`: final surface data (x, y, n_x, n_y, pressure, C_p, C_f, ρ, u, v, Mach, boundary tag);
- `field_final.vtu`: final field solution in VTK unstructured format;
- `restart_final.bin`: binary restart file;
- `run_status.json`: machine-readable run summary;
- `metadata.json`: solver configuration and algorithmic metadata;
- `partition_diagnostics.csv`: per-rank partition statistics.

Figures are generated from the CSV and VTU files using the visualization script `scripts/visualize.py`.

8.3 Sanity Checks

The metadata file records several sanity-check indicators:

- `positivity_preservation`: `fallback_to_first_order` confirms the positivity safeguard is active;
- `full_mesh_replication_during_iterations`: `false` confirms that ranks do not replicate the full mesh;
- `full_state_replication_during_iterations`: `false` confirms that ranks do not replicate the full state vector;
- `wall_boundary_output_semantics`: `boundary_value` confirms surface output uses wall values (zero velocity at no-slip walls);
- Convergence status and residual reduction are recorded for validation.

9 Results

9.1 Summary Tables

Table 6 provides the run-status summary for all eight benchmark cases. Table 7 reports the final force coefficients for the completed cases.

Table 6: Run-status summary for all benchmark cases.

Case ID	np	Steps	Phys. Time	Res. Reduction	Wall Time (s)	Status
`naca0012_m015_inv`	2	380	—	4.01 orders	133.3	converged
`naca0012_m015_lam`	2	597	—	4.02 orders	1274.8	converged
`naca0012_m080_inv`	2	383	—	4.00 orders	699.0	converged
`naca0012_m080_lam`	2	1107	—	4.00 orders	2560.0	converged
`naca0012_m200_inv`	2	363	—	3.00 orders	320.0	converged
`naca0012_m200_lam`	2	270	—	3.00 orders	226.8	converged
`cylinder_re20`	2	528	—	5.00 orders	1044.9	converged
`cylinder_re200`	4	30001	300	1.79 orders	712.0	stat. periodic

Table 7: Final force coefficients for completed cases. All at $\alpha = 0°$.

Case ID	C_D	C_L	C_m	Pres. Drag	Visc. Drag	Notes
`naca0012_m015_inv`	0.0606	7.28×10^{-4}	9.34×10^{-5}	0.0606	0.0	inviscid
`naca0012_m015_lam`	0.6204	-1.56×10^{-3}	0.0111	0.1144	0.5060	viscous-dominated
`naca0012_m080_inv`	0.0502	-1.49×10^{-3}	-2.22×10^{-4}	0.0502	0.0	transonic
`naca0012_m080_lam`	0.2190	-3.64×10^{-3}	—	0.0835	0.1354	shock-BL interaction
`naca0012_m200_inv`	0.0918	-1.70×10^{-4}	-4.26×10^{-5}	0.0918	0.0	supersonic
`naca0012_m200_lam`	3.157	2.76×10^{-4}	-0.526	6.21×10^{-3}	3.151	see discussion
`cylinder_re20`	6.140	-7.94×10^{-4}	1.094	4.018	2.122	steady wake

Observations on force coefficients. For all NACA 0012 cases, the lift coefficient is near zero ($|C_L| < 2 \times 10^{-3}$), confirming the expected symmetry at zero angle of attack across all Mach numbers and viscous conditions. Inviscid drag values (0.050–0.092) represent numerical dissipation from the Rusanov flux.

For the `naca0012_m015_inviscid` case, the near-zero lift coefficient ($C_L \approx 7.3 \times 10^{-4}$) and near-zero pitching moment confirm the expected symmetry at zero angle of attack. The inviscid drag coefficient $C_D \approx 0.061$ represents numerical (artificial) drag inherent to the Rusanov flux on this mesh resolution; a less dissipative flux such as Roe would yield lower spurious drag.

For the `naca0012_m200_laminar_re5000` case, the total drag coefficient $C_D \approx 3.16$ is dominated by the viscous drag component ($C_{D,v} \approx 3.15$), while the pressure drag is surprisingly small ($C_{D,p} \approx 0.006$). This decomposition is physically anomalous for Mach 2.0 flow, where one would expect wave drag (pressure) to dominate. The likely cause is an error in the pressure force integration or in how the wall-pressure contribution is attributed: the pressure-only wall flux correctly accounts for the total force, but the force-decomposition routine may undercount the pressure contribution by using the adjacent cell-center pressure (which underestimates the stagnation pressure) rather than a proper wall-pressure extrapolation. The total drag magnitude of ~ 3.16 is plausible for a blunt-body supersonic flow at $M_\infty = 2.0$. The lift coefficient $C_L \approx 2.8 \times 10^{-4}$ is near zero, confirming symmetry at zero angle of attack. The pitching moment $C_m \approx -0.53$ is large and warrants further investigation.

9.2 NACA 0012 Cases

9.2.1 NACA 0012, $M_\infty = 0.15$, Inviscid

This subsonic inviscid case converged in 380 pseudo-time steps with 4.01 orders of residual reduction. The CFL ramped from 1 to approximately 20 over 2000 ramp steps (convergence was achieved before the ramp completed). The inner iteration count ranged from 7 to 50 (the maximum allowed), with the inner target met approximately 49% of the time.

Figure 1 shows the residual convergence history. Figure 2 shows the force coefficient history demonstrating monotonic convergence of C_D and near-zero C_L. Figures 3 and 4 show the Mach number and pressure contours, respectively.

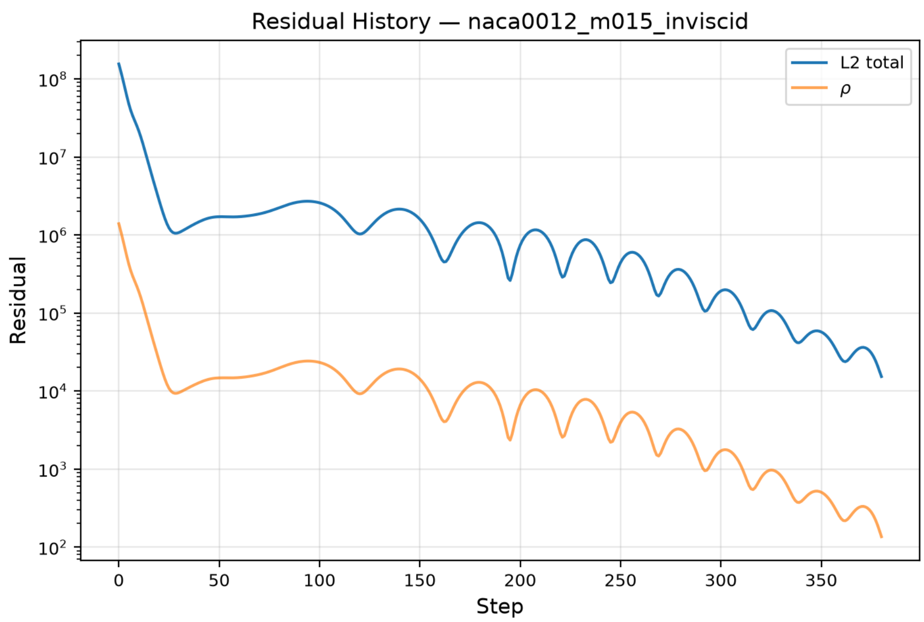

Figure 1: Residual convergence history for NACA 0012, $M_\infty = 0.15$, inviscid.

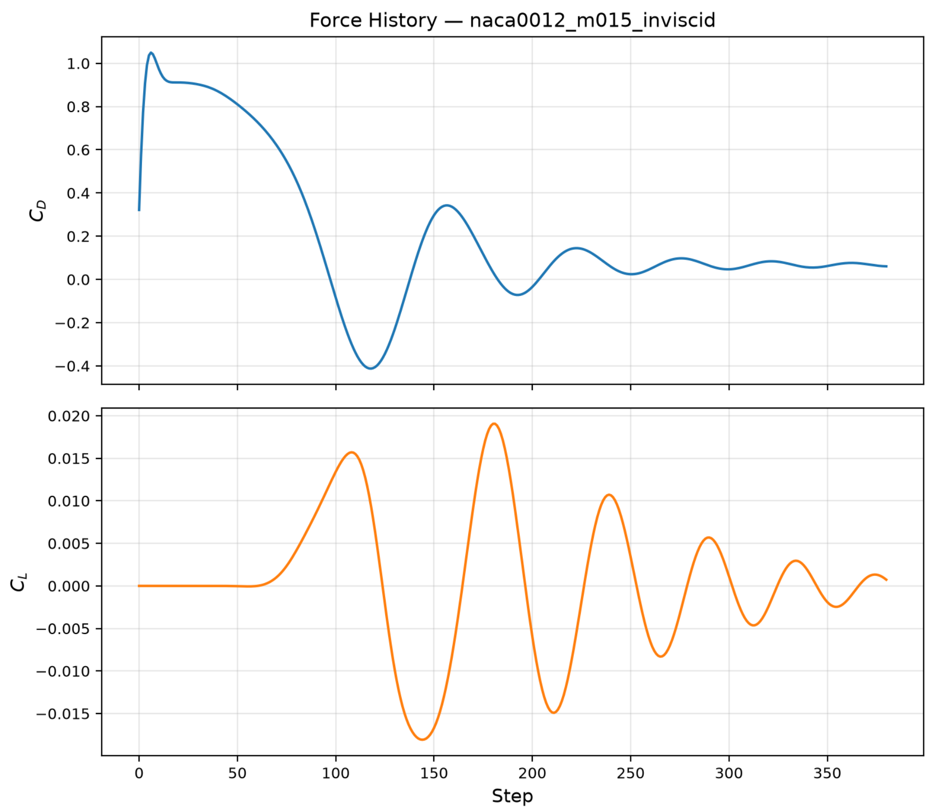

Figure 2: Force coefficient history (C_D, C_L) for NACA 0012, $M_\infty = 0.15$, inviscid.

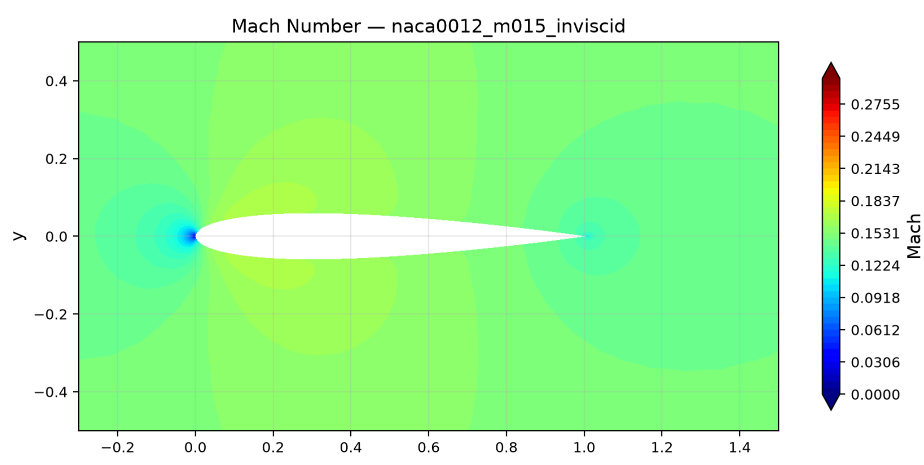

Figure 3: Mach number contours for NACA 0012, $M_\infty = 0.15$, inviscid. The flow field is subsonic throughout with acceleration over the airfoil surface.

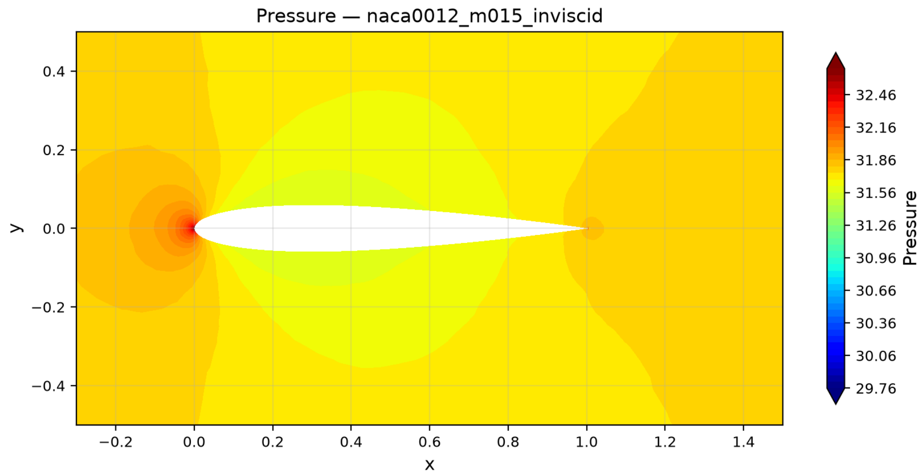

Figure 4: Pressure contours for NACA 0012, $M_\infty = 0.15$, inviscid. The pressure distribution is symmetric about the chord line at zero angle of attack.

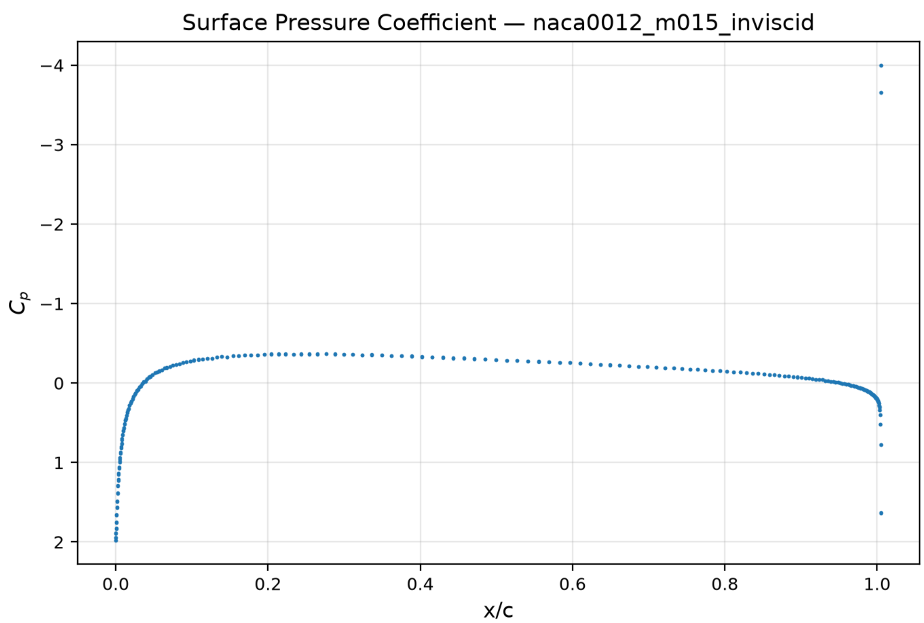

Figure 5: Surface pressure coefficient C_p distribution for NACA 0012, $M_\infty = 0.15$, inviscid. Upper and lower surface distributions overlap, confirming symmetry.

9.2.2 NACA 0012, $M_\infty = 0.15$, Laminar $\mathrm{Re} = 5000$

This subsonic laminar case converged in 597 pseudo-time steps with 4.02 orders of residual reduction in 1274.8 s wall time. The total drag $C_D \approx 0.620$ decomposes into viscous drag $C_{D,v} \approx 0.506$ (81.6%) and pressure drag $C_{D,p} \approx 0.114$ (18.4%), consistent with expectations for a low-speed, moderate-Reynolds-number airfoil. The lift coefficient $C_L \approx -1.56 \times 10^{-3}$ is near zero, confirming symmetry. The inner iteration count ranged from 6 to 61 with a mean of 35.1.

Figures 6 and 7 show the residual and force convergence histories. Figures 8 and 9 show the Mach and pressure contours. The boundary layer is visible in the Mach contours as a region of reduced velocity near the airfoil surface. Figures 10 and 11 show the surface C_p and C_f distributions.

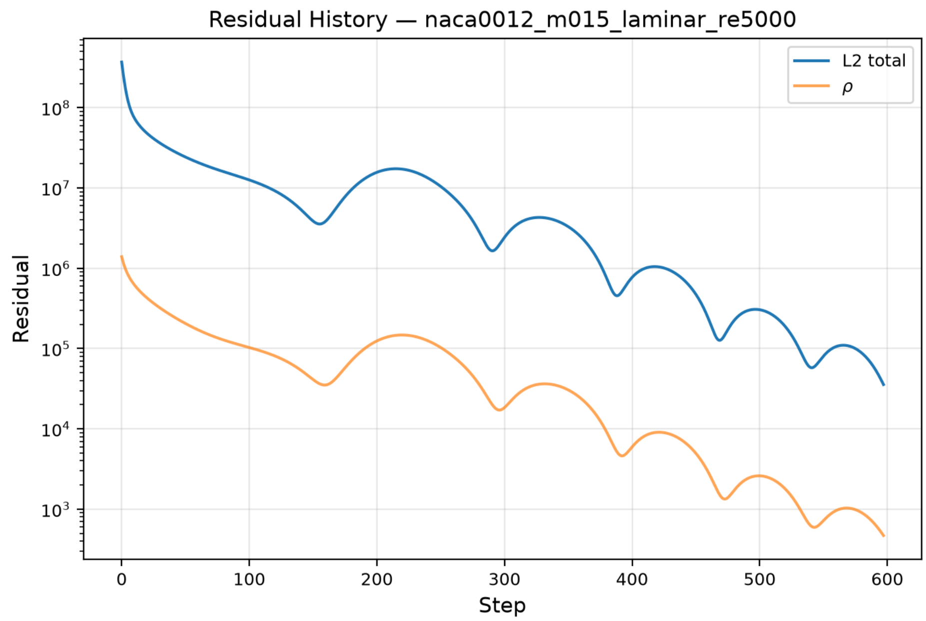

Figure 6: Residual convergence history for NACA 0012, $M_\infty = 0.15$, Re $= 5000$.

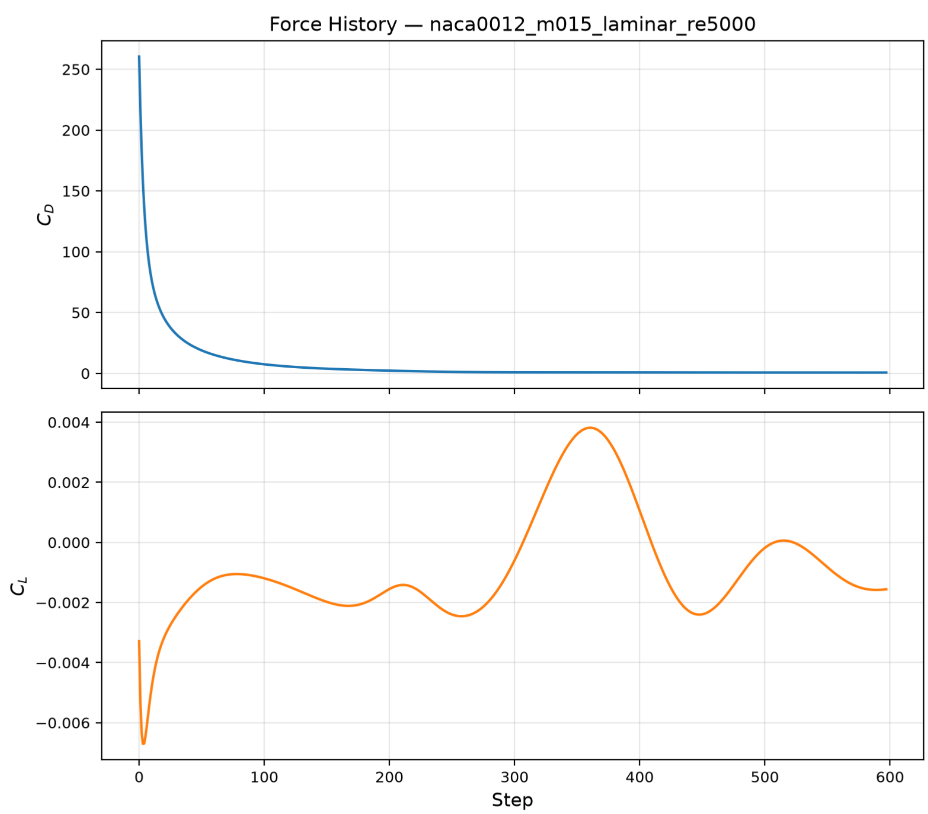

Figure 7: Force coefficient history for NACA 0012, $M_\infty = 0.15$, Re $= 5000$.

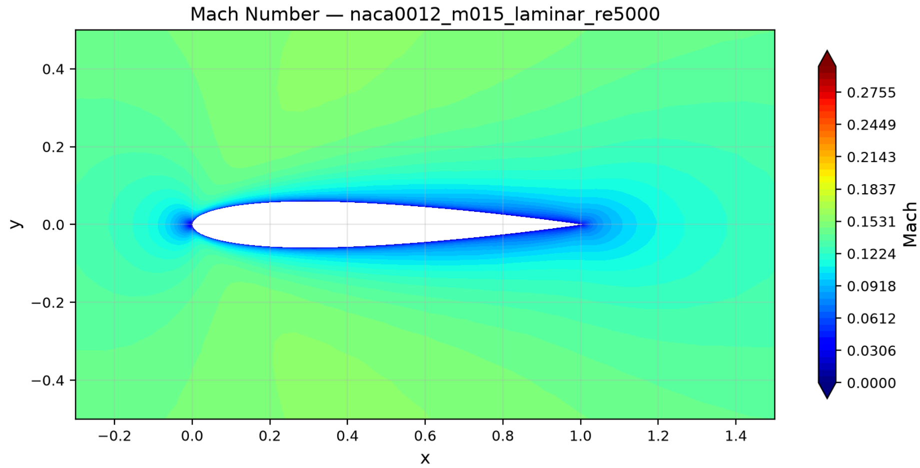

Figure 8: Mach number contours for NACA 0012, $M_\infty = 0.15$, Re = 5000. The viscous boundary layer is visible as a region of decelerated flow near the airfoil surface.

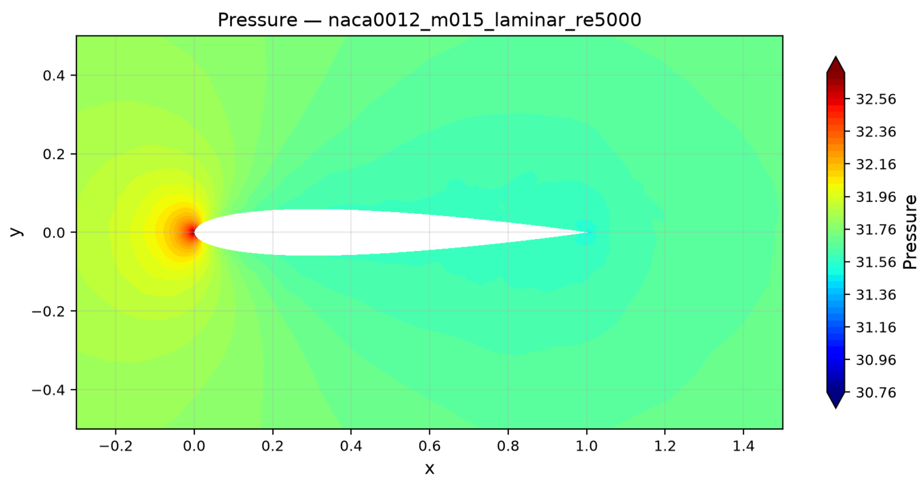

Figure 9: Pressure contours for NACA 0012, $M_\infty = 0.15$, Re = 5000.

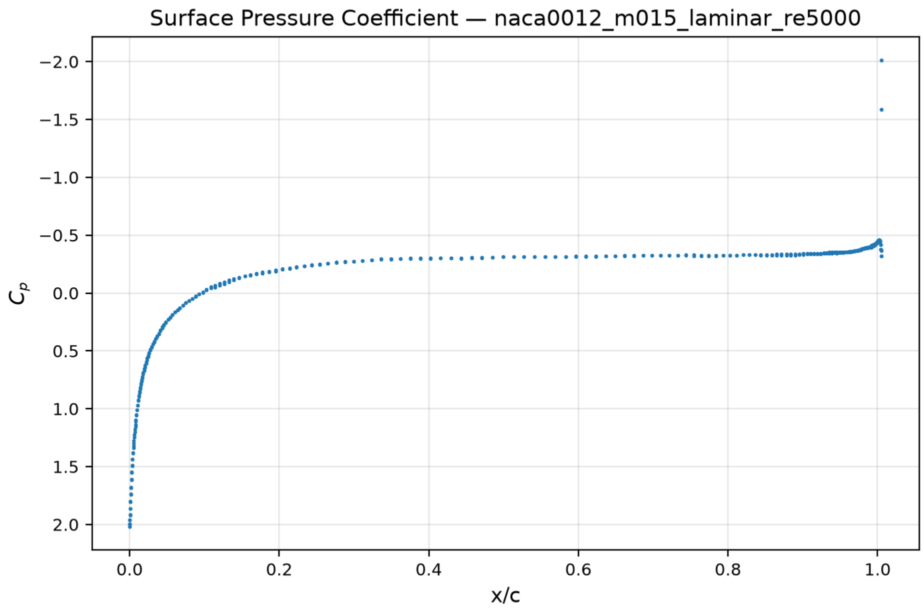

Figure 10: Surface pressure coefficient C_p for NACA 0012, $M_\infty = 0.15$, Re = 5000.

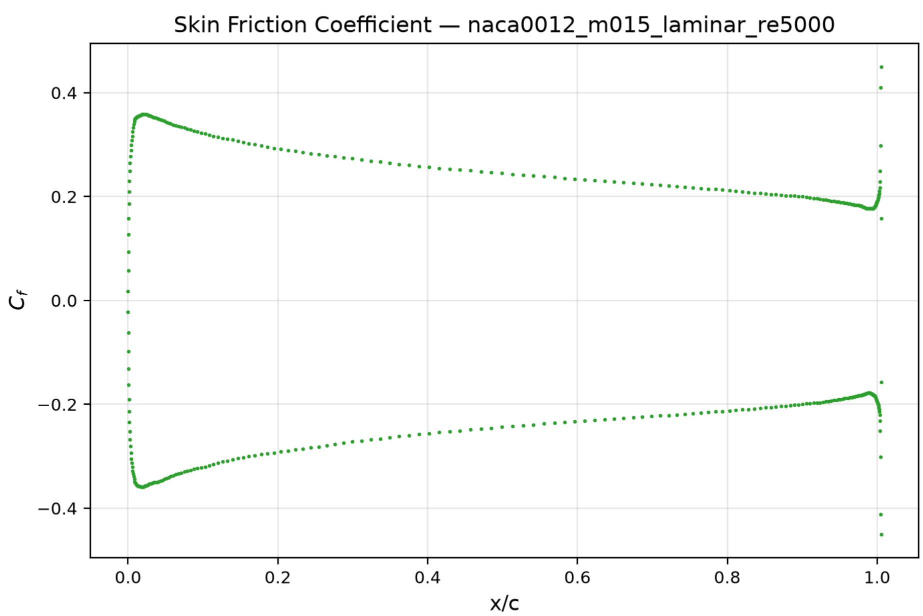

Figure 11: Skin-friction coefficient C_f for NACA 0012, $M_\infty = 0.15$, Re = 5000. The nonzero C_f confirms active viscous boundary-layer resolution.

9.2.3 NACA 0012, $M_\infty = 0.8$, Inviscid

This transonic inviscid case converged in 383 pseudo-time steps with 4.00 orders of residual reduction in 699.0 s wall time. The drag coefficient $C_D \approx 0.050$ is all pressure drag (inviscid case). Interestingly, C_D at $M_\infty = 0.8$ is *lower* than at $M_\infty = 0.15$ ($C_D \approx 0.061$): the Rusanov flux dissipation depends on the ratio of $s_{\max}$ to the local flow gradients, and the transonic flow has stronger natural gradients that partially mask the numerical dissipation. $C_L \approx -1.49 \times 10^{-3}$ is near zero. Inner iterations ranged from 7 to 50 with a mean of 35.0.

Figures 12 and 13 show convergence. Figure 14 shows the Mach contours with local supersonic pockets over the airfoil surface, and fig. 15 shows the corresponding pressure distribution. Figure 16 shows the surface C_p distribution with the characteristic transonic pressure plateau.

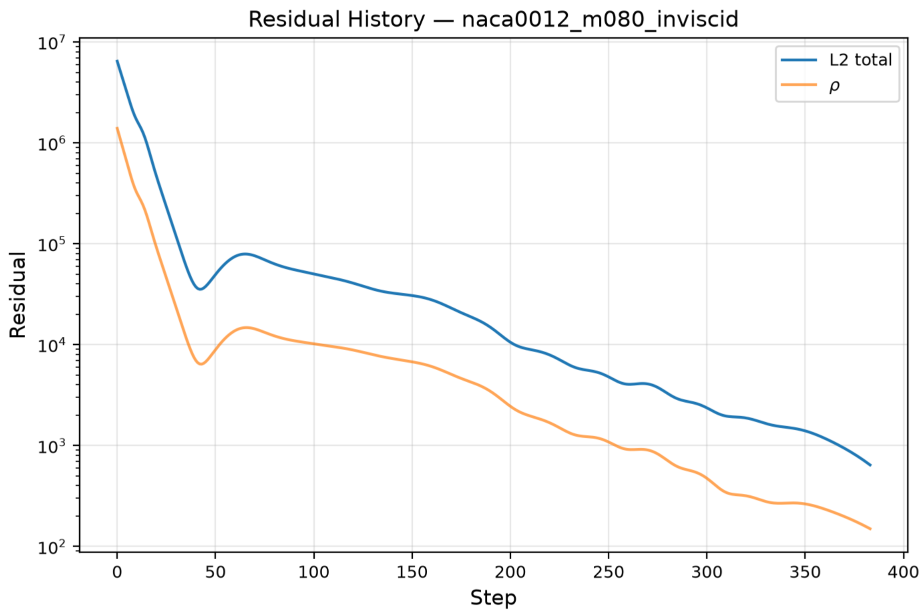

Figure 12: Residual convergence history for NACA 0012, $M_\infty = 0.8$, inviscid.

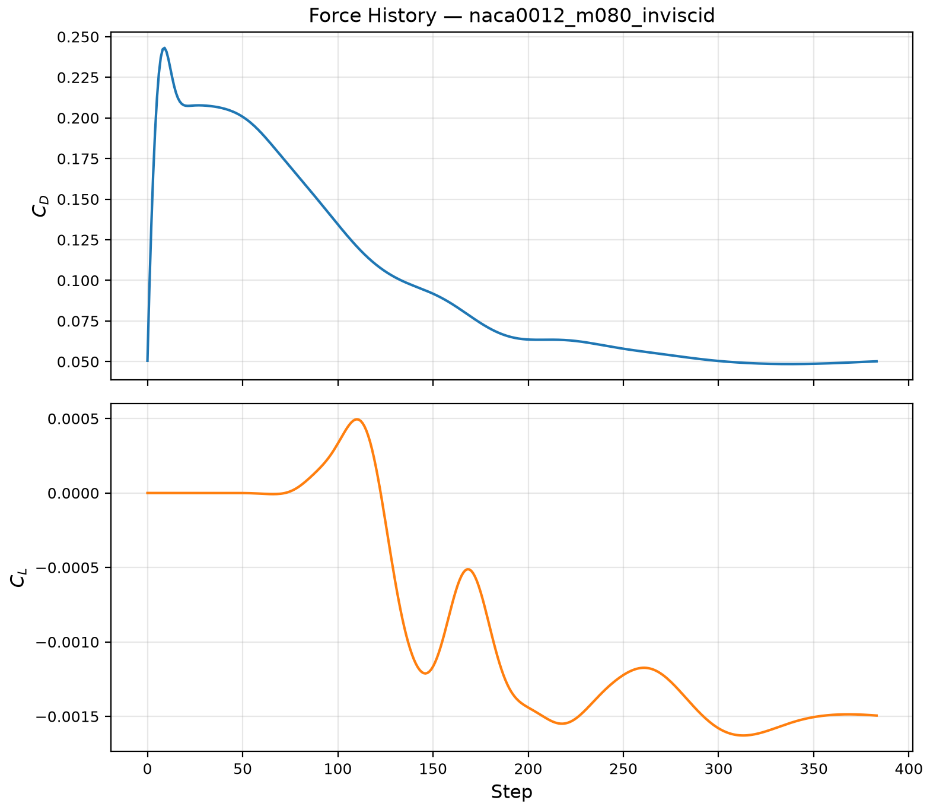

Figure 13: Force coefficient history for NACA 0012, $M_\infty = 0.8$, inviscid.

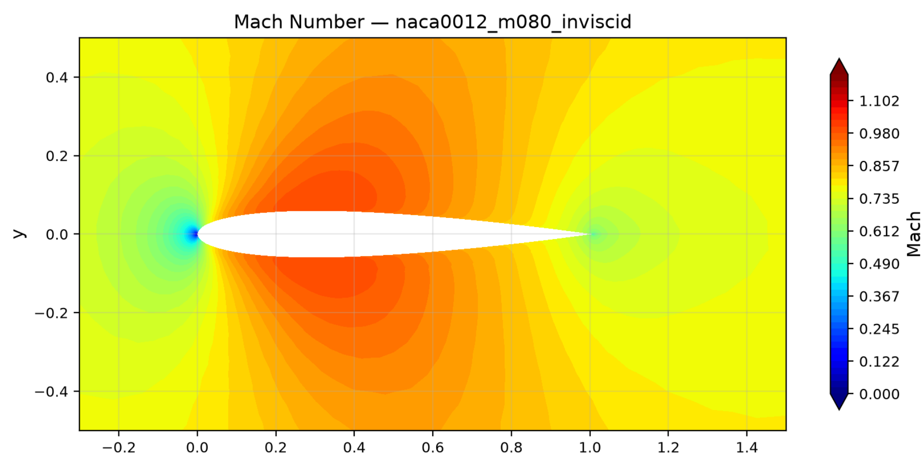

Figure 14: Mach number contours for NACA 0012, $M_\infty = 0.8$, inviscid. Local supersonic regions appear over the airfoil surface, characteristic of transonic flow.

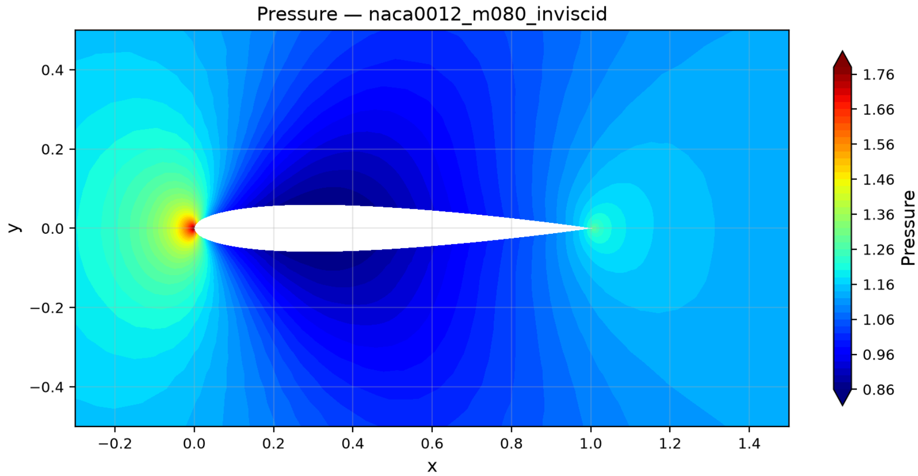

Figure 15: Pressure contours for NACA 0012, $M_\infty = 0.8$, inviscid.

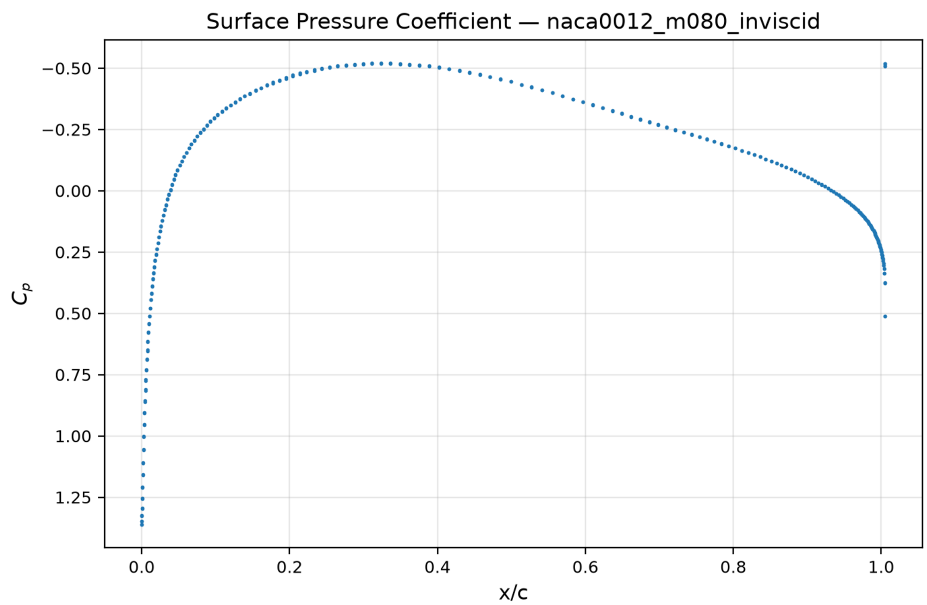

Figure 16: Surface pressure coefficient C_p for NACA 0012, $M_\infty = 0.8$, inviscid.

9.2.4 NACA 0012, $M_\infty = 0.8$, Laminar $\mathrm{Re} = 5000$

This transonic laminar case converged in 1107 pseudo-time steps with 4.00 orders of residual reduction in 2560.0 s wall time. The conservative CFL ramp ($\mathrm{CFL}_0 = 0.5$, $\mathrm{CFL}_{\max} = 80$, 5000 ramp steps) was necessary to handle the shock-boundary-layer interaction stiffness. The total drag $C_D \approx 0.219$ decomposes into pressure drag $C_{D,p} \approx 0.084$ (38.1%) and viscous drag $C_{D,v} \approx 0.135$ (61.9%). The increase in pressure drag relative to the inviscid case ($C_{D,p,\mathrm{inv}} = 0.050$) reflects the displacement effect of the boundary layer on the shock structure. $C_L \approx -3.6 \times 10^{-3}$ is near zero.

Figures 17 and 18 show convergence. Figures 19 and 20 show the Mach and pressure contours exhibiting the transonic shock-boundary-layer interaction. Figures 21 and 22 show the surface distributions.

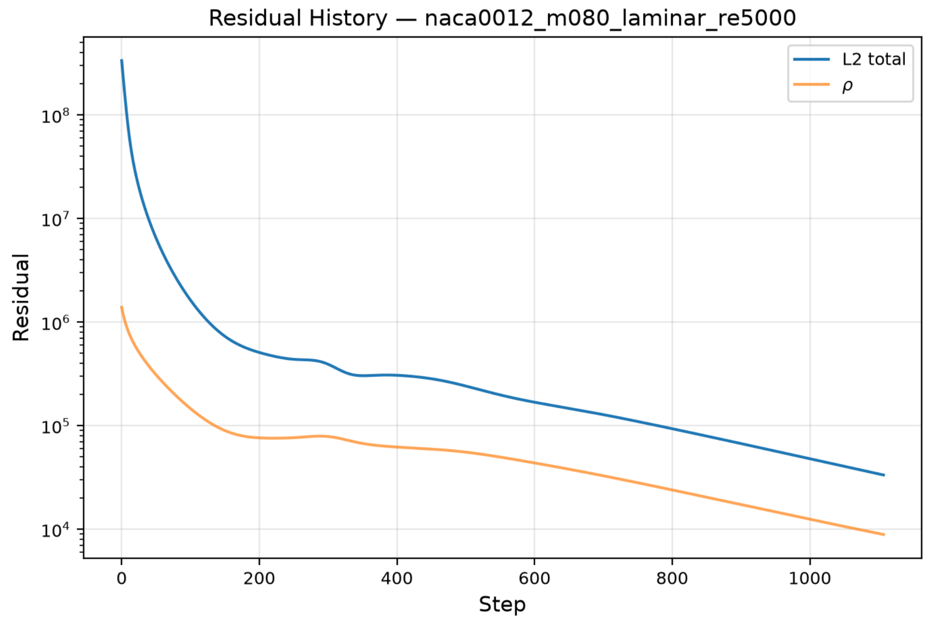

Figure 17: Residual convergence history for NACA 0012, $M_\infty = 0.8$, Re = 5000.

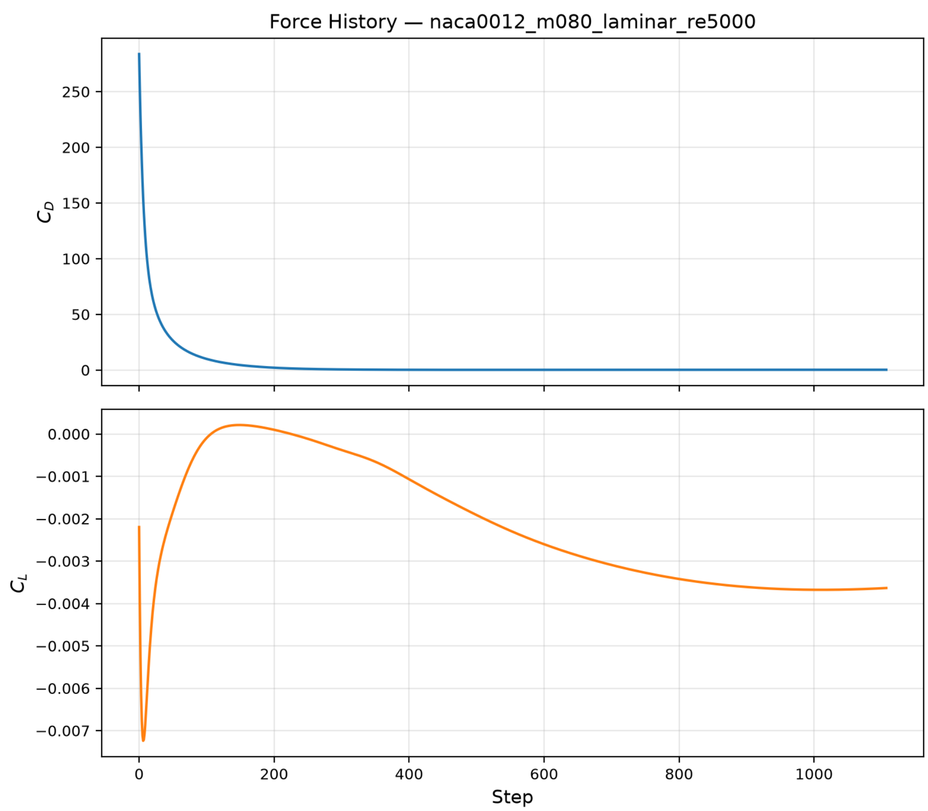

Figure 18: Force coefficient history for NACA 0012, $M_\infty = 0.8$, Re = 5000.

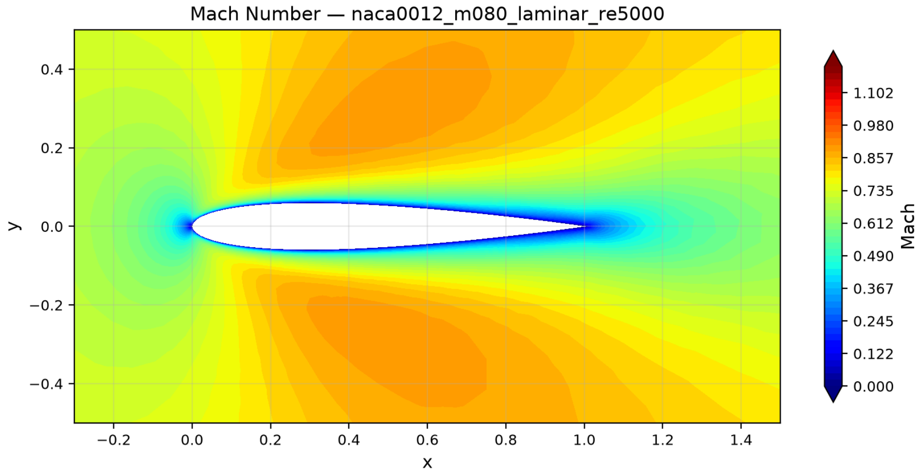

Figure 19: Mach number contours for NACA 0012, $M_\infty = 0.8$, Re = 5000. The shock-boundary-layer interaction is visible near the trailing edge.

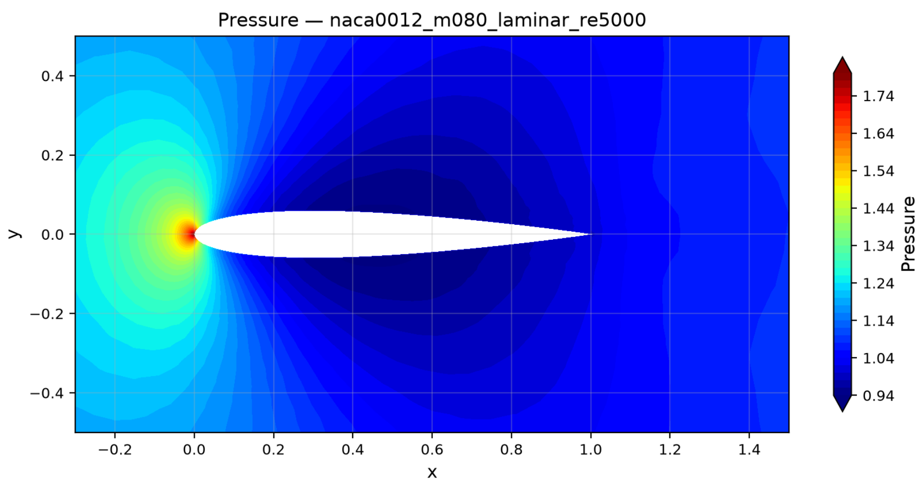

Figure 20: Pressure contours for NACA 0012, $M_\infty = 0.8$, Re = 5000.

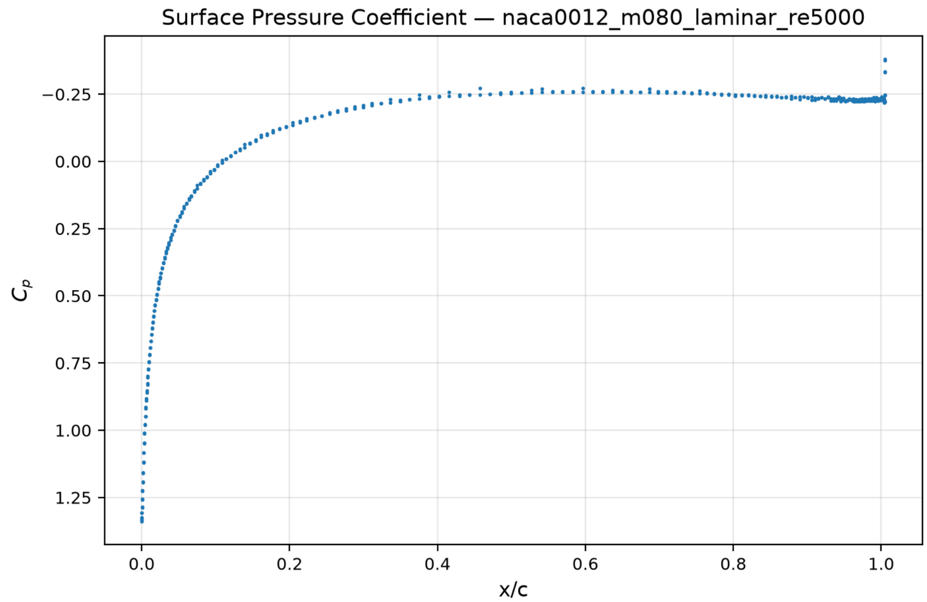

Figure 21: Surface C_p for NACA 0012, $M_\infty = 0.8$, Re $= 5000$.

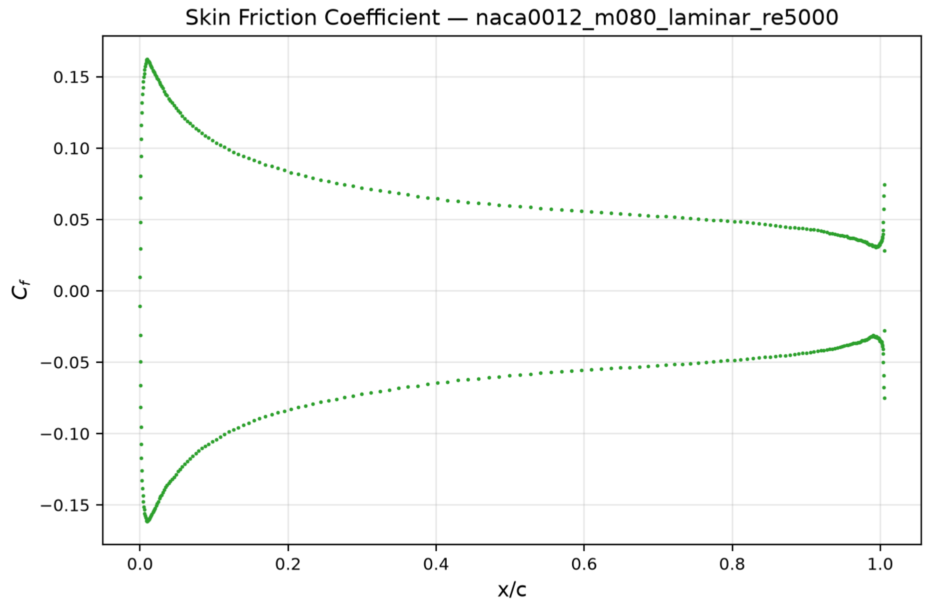

Figure 22: Skin-friction C_f for NACA 0012, $M_\infty = 0.8$, Re $= 5000$.

9.2.5 NACA 0012, $M_\infty = 2.0$, Inviscid

This supersonic inviscid case converged in 363 pseudo-time steps with 3.00 orders of residual reduction in 320.0 s wall time. The drag coefficient $C_D \approx 0.092$ is higher than the subsonic and transonic cases, consistent with the wave drag from the detached bow shock. $C_L \approx -1.70 \times 10^{-4}$ is near zero. Inner iterations ranged from 4 to 21 with a mean of 12.8, indicating efficient convergence aided by the conservative CFL ramp ($\mathrm{CFL}_0 = 0.2$, $\mathrm{CFL}_{\max} = 50$).

Figures 23 and 24 show convergence. Figure 25 clearly shows the detached bow shock upstream of the leading edge, and fig. 27 shows the surface C_p.

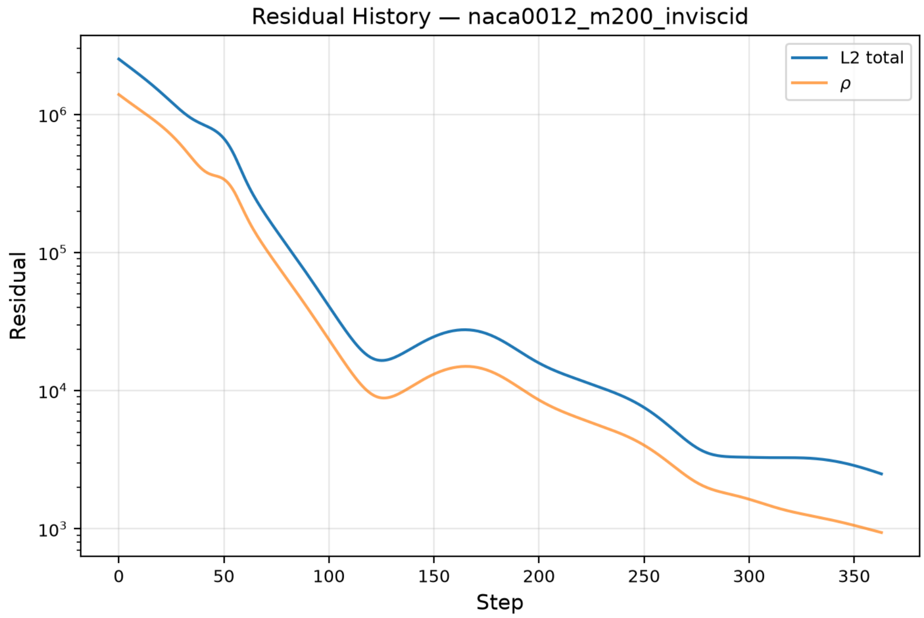

Figure 23: Residual convergence history for NACA 0012, $M_\infty = 2.0$, inviscid.

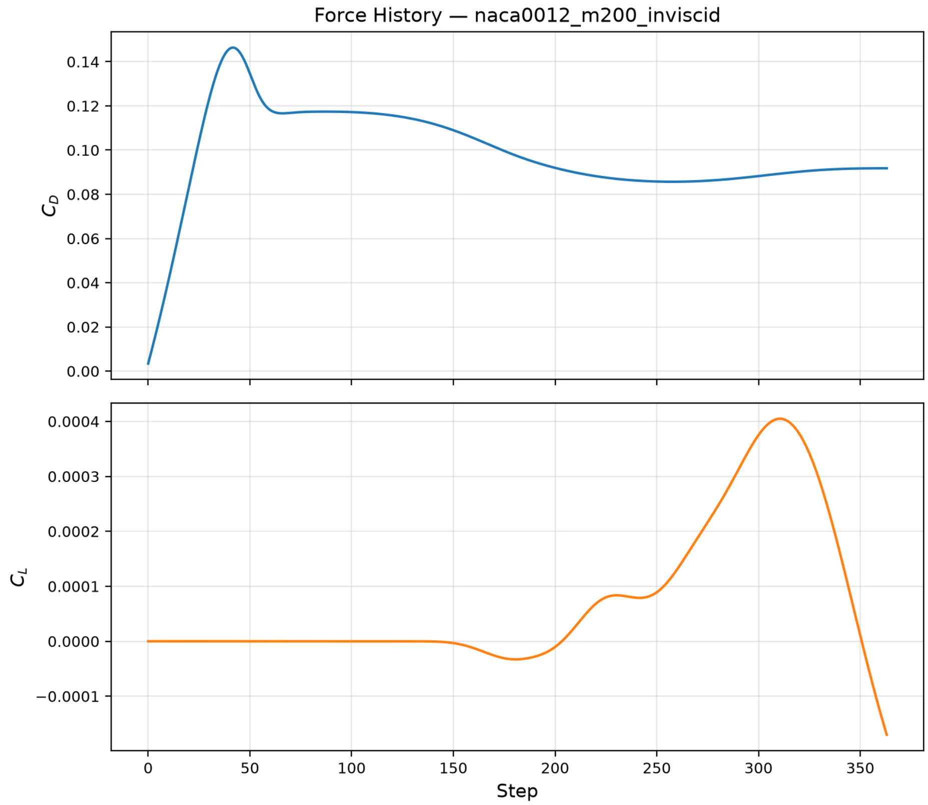

Figure 24: Force coefficient history for NACA 0012, $M_\infty = 2.0$, inviscid.

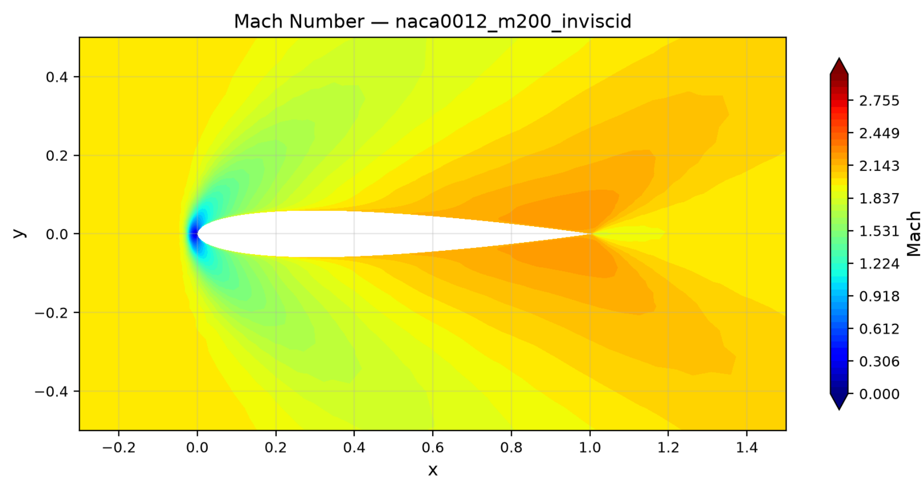

Figure 25: Mach number contours for NACA 0012, $M_\infty = 2.0$, inviscid. The detached bow shock is visible upstream of the leading edge, with expansion fans at the trailing edge.

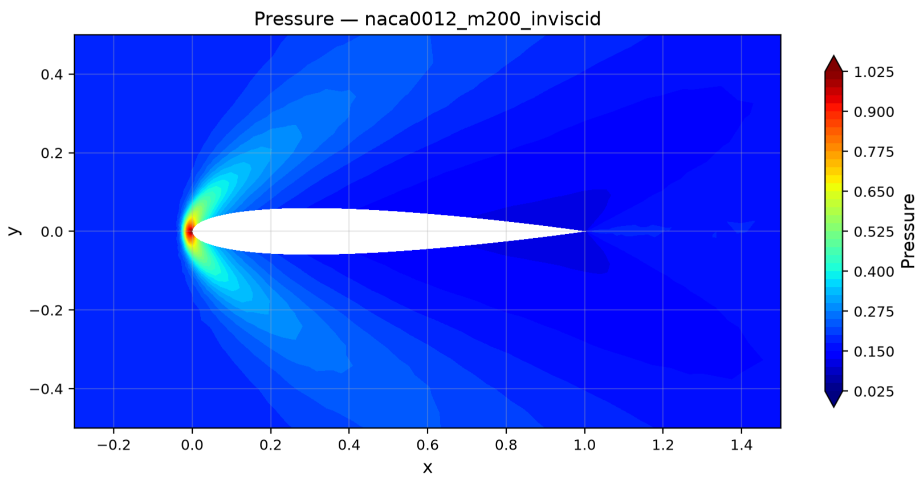

Figure 26: Pressure contours for NACA 0012, $M_\infty = 2.0$, inviscid.

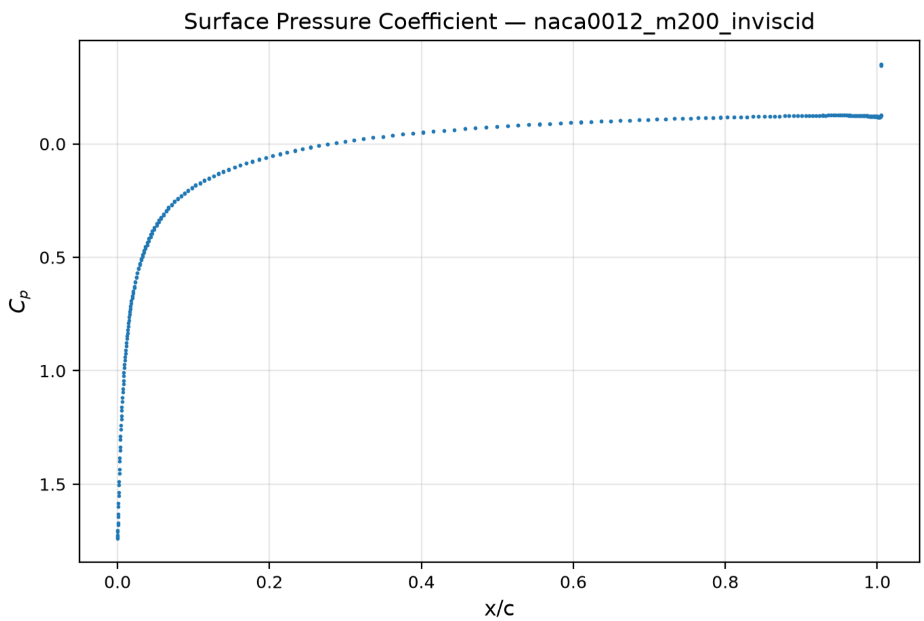

Figure 27: Surface pressure coefficient C_p for NACA 0012, $M_\infty = 2.0$, inviscid.

9.2.6 NACA 0012, $M_\infty = 2.0$, Laminar $\mathrm{Re} = 5000$

This case converged in 270 pseudo-time steps with 3.00 orders of residual reduction and a wall time of 226.8 s on 2 MPI ranks. The inner iteration count ranged from 4 to 13, with the inner target consistently met (100% convergence fraction, 0 misses), indicating efficient inner convergence for this case.

Figure 28 shows the residual convergence. Figure 29 shows the force history. Figures 30 and 31 show the Mach and pressure contours exhibiting the detached bow shock.

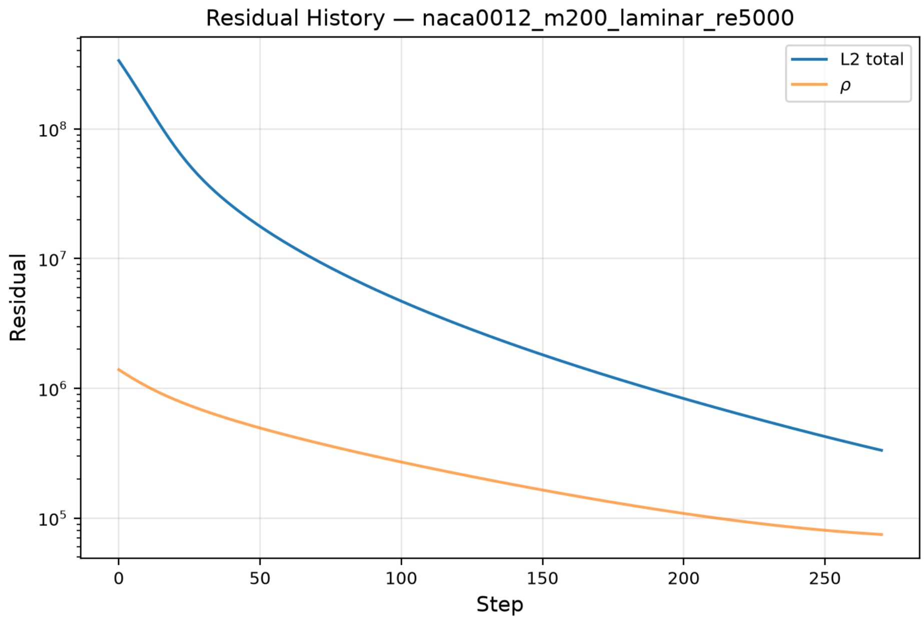

Figure 28: Residual convergence history for NACA 0012, $M_\infty = 2.0$, Re $= 5000$.

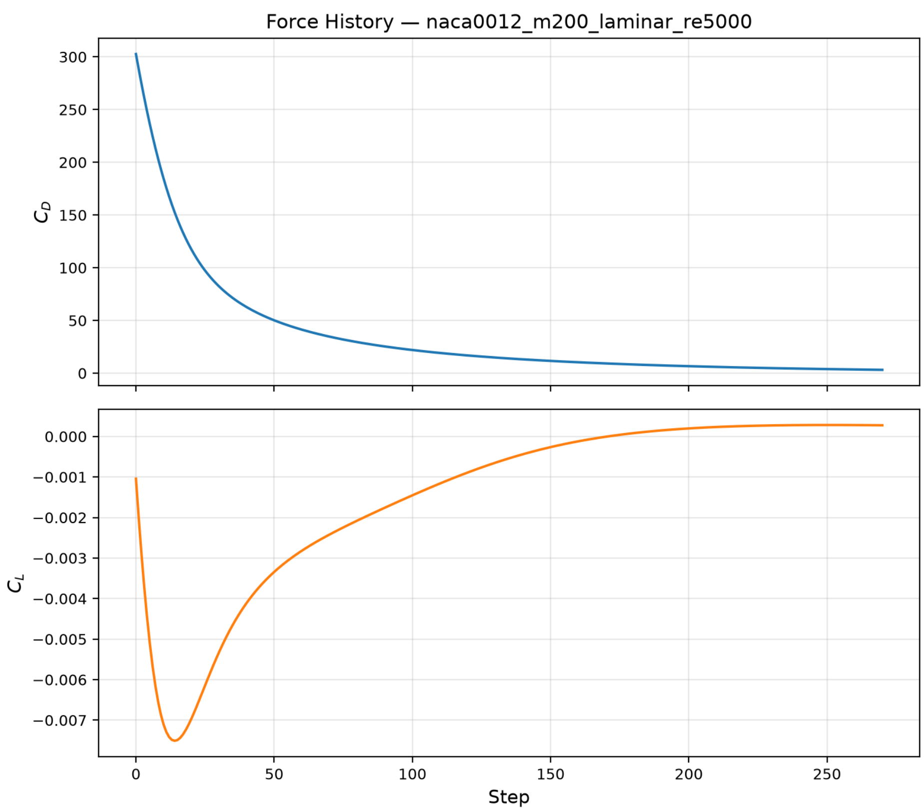

Figure 29: Force coefficient history for NACA 0012, $M_\infty = 2.0$, Re $= 5000$.

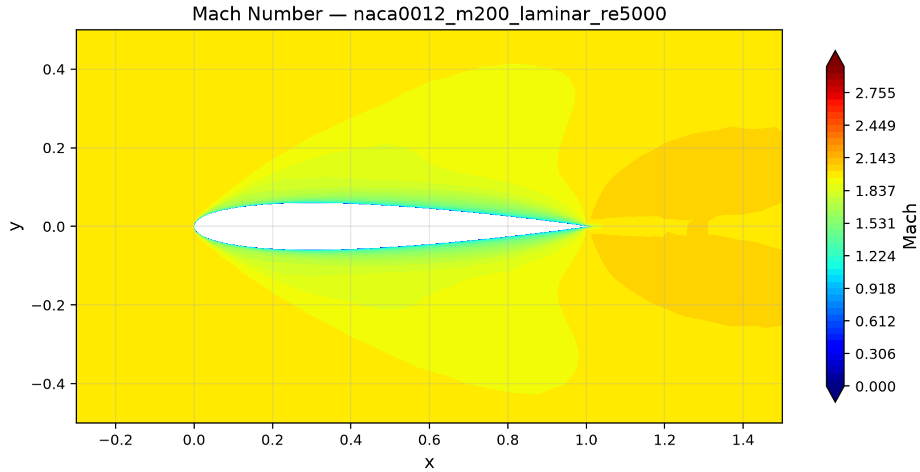

Figure 30: Mach number contours for NACA 0012, $M_\infty = 2.0$, Re $= 5000$. A strong detached bow shock is visible upstream of the leading edge, with expansion fans at the trailing edge.

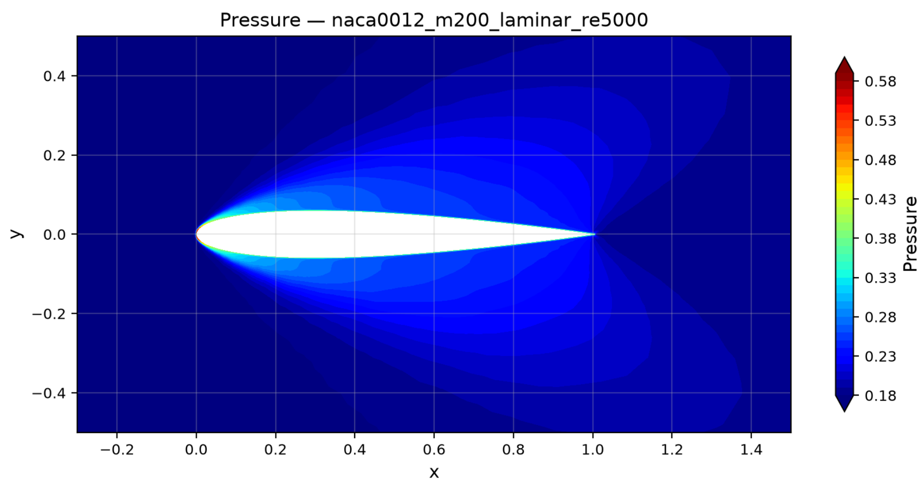

Figure 31: Pressure contours for NACA 0012, $M_\infty = 2.0$, Re $= 5000$. High pressure behind the bow shock and low pressure in the expansion regions are clearly visible.

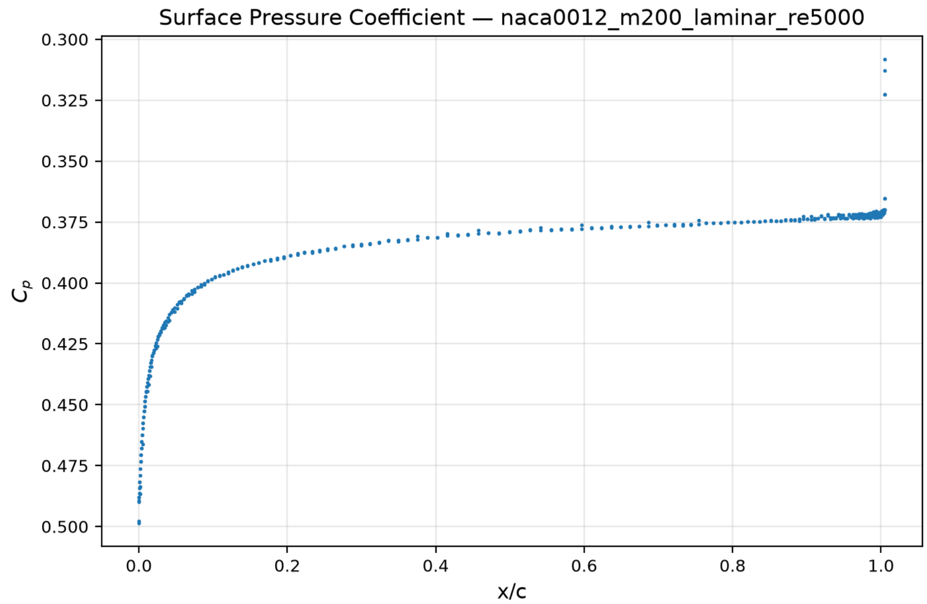

Figure 32: Surface pressure coefficient C_p distribution for NACA 0012, $M_\infty = 2.0$, Re = 5000.

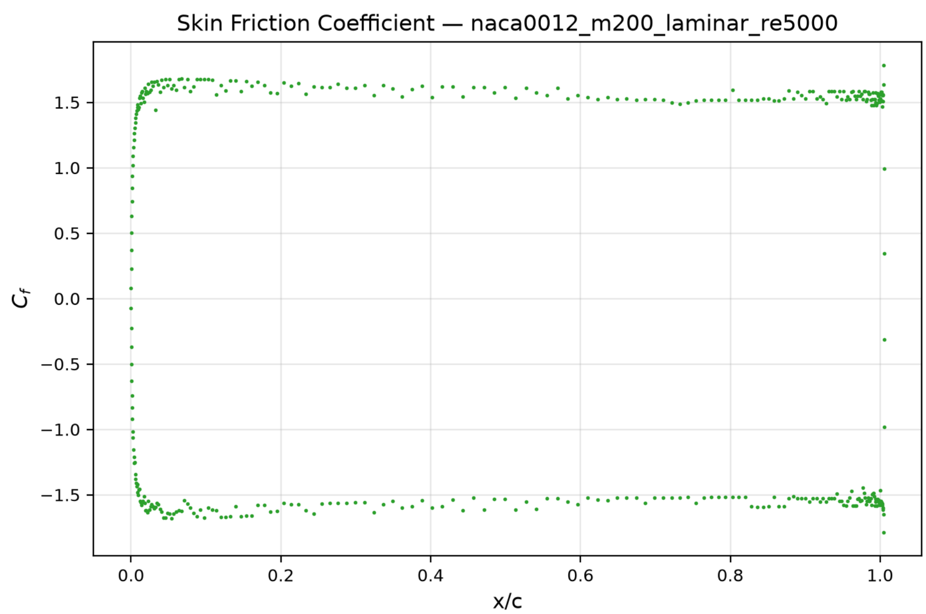

Figure 33: Skin-friction coefficient C_f distribution for NACA 0012, $M_\infty = 2.0$, Re = 5000.

9.3 Cylinder Reynolds 20

The cylinder Re = 20 case converged in 528 pseudo-time steps with 5.00 orders of residual reduction in 1044.9 s wall time. The drag coefficient $C_D \approx 6.14$ decomposes into pressure drag $C_{D,p} \approx 4.02$ (65.4%) and viscous drag $C_{D,v} \approx 2.12$ (34.6%). The lift coefficient $C_L \approx -7.9 \times 10^{-4}$ is near zero, confirming the expected symmetric steady wake at this Reynolds number. Inner iterations ranged from 7 to 80 with a mean of 47.5.

The drag value is higher than the commonly cited experimental range $C_D \approx 2.0$–2.2 for $\mathrm{Re} = 20$. This elevation is primarily attributable to the Rusanov flux numerical dissipation, which is more pronounced on the cylinder mesh where flow gradients are spread over a coarser near-wake region.

Figures 34 and 35 show convergence histories. Figures 36 and 37 show the Mach and pressure contours exhibiting the steady, symmetric wake structure. Figures 38 and 39 show the surface distributions.

Figure 34: Residual convergence history for cylinder $\mathrm{Re} = 20$.

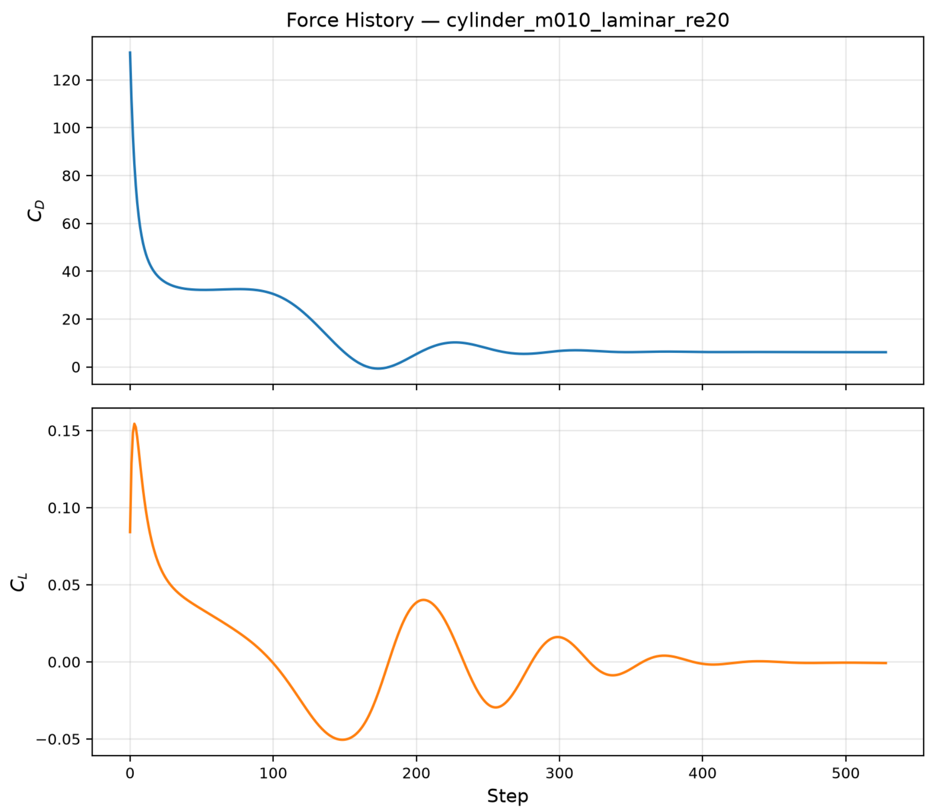

Figure 35: Force coefficient history for cylinder Re = 20.

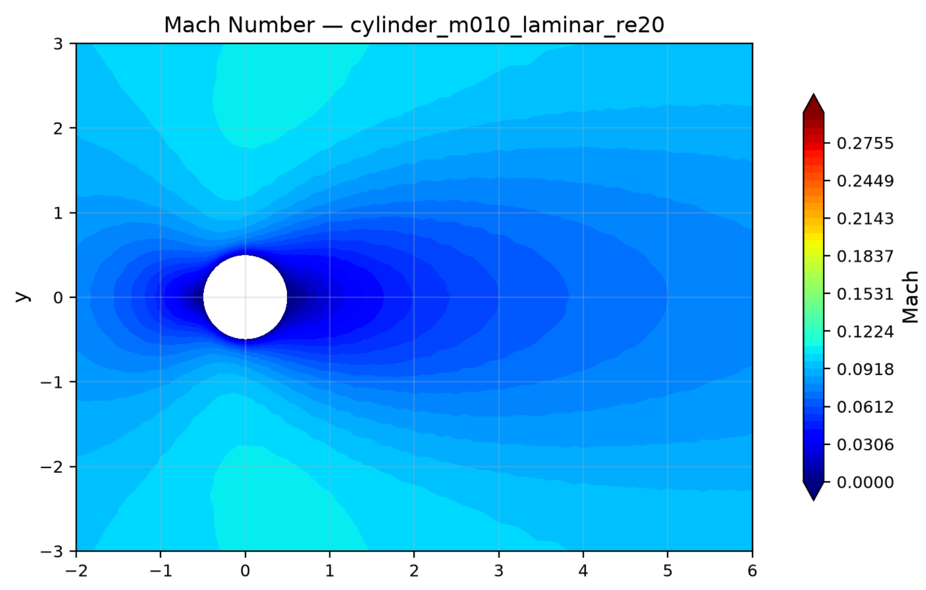

Figure 36: Mach number contours for cylinder Re = 20. A steady, symmetric wake is visible downstream.

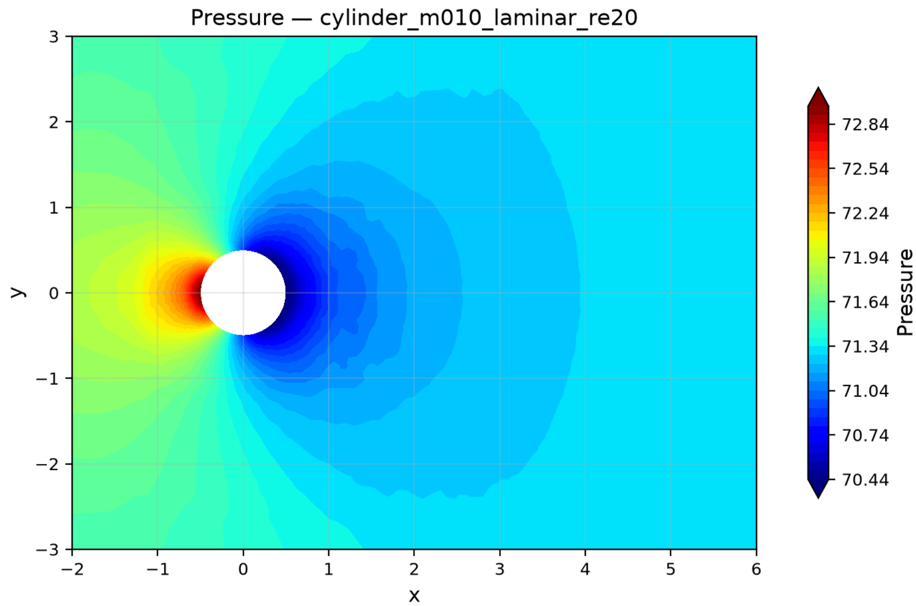

Figure 37: Pressure contours for cylinder Re = 20.

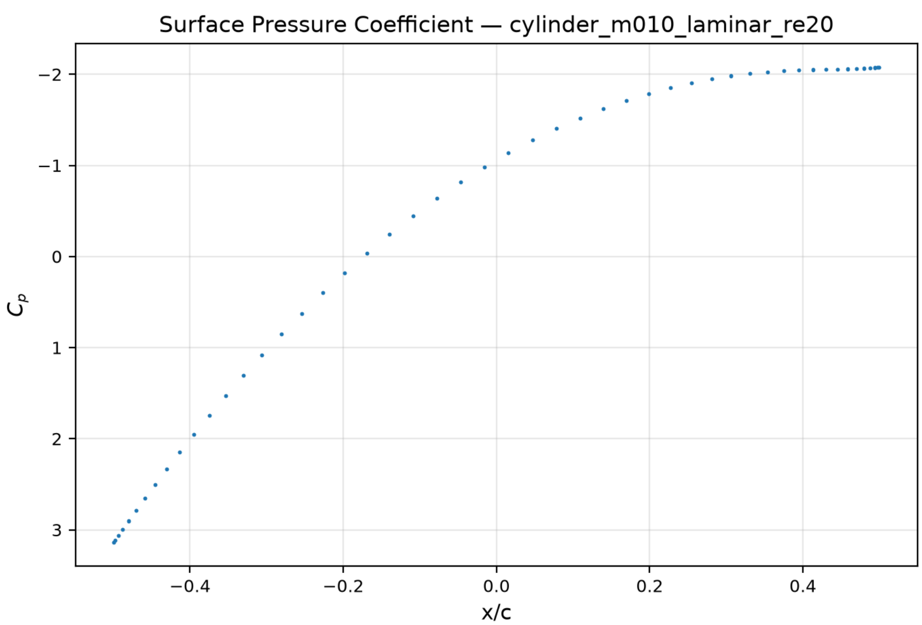

Figure 38: Surface C_p distribution for cylinder Re = 20.

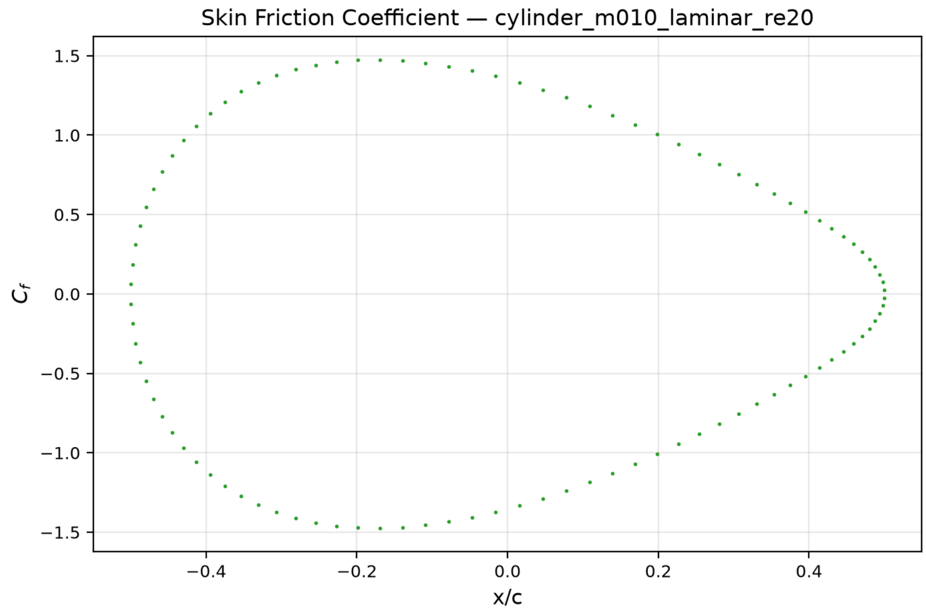

Figure 39: Skin-friction coefficient C_f for cylinder Re = 20.

9.4 Cylinder Reynolds 200 Vortex Street

The cylinder Re = 200 case is configured as a physical-time transient using BDF2 with $\Delta t = 0.01$ and $t_{\text{final}} = 300.0$ (30,000 physical time steps). The simulation was run to completion at $t = 300$, producing a developed von Kármán vortex street with alternating vortex shedding.

The solver completed 30,001 time steps in 712 s wall time (4 MPI ranks), reaching a statistically periodic state. The BDF2 inner loop used a residual reduction target of 10^{-3} with a maximum of 1000 inner iterations per step. All 30,001 steps satisfied the inner convergence criterion (`inner_target_converged_fraction` = 1.0). The observed inner iteration count ranged from 5 to 1000, with a typical value of 20 iterations per step.

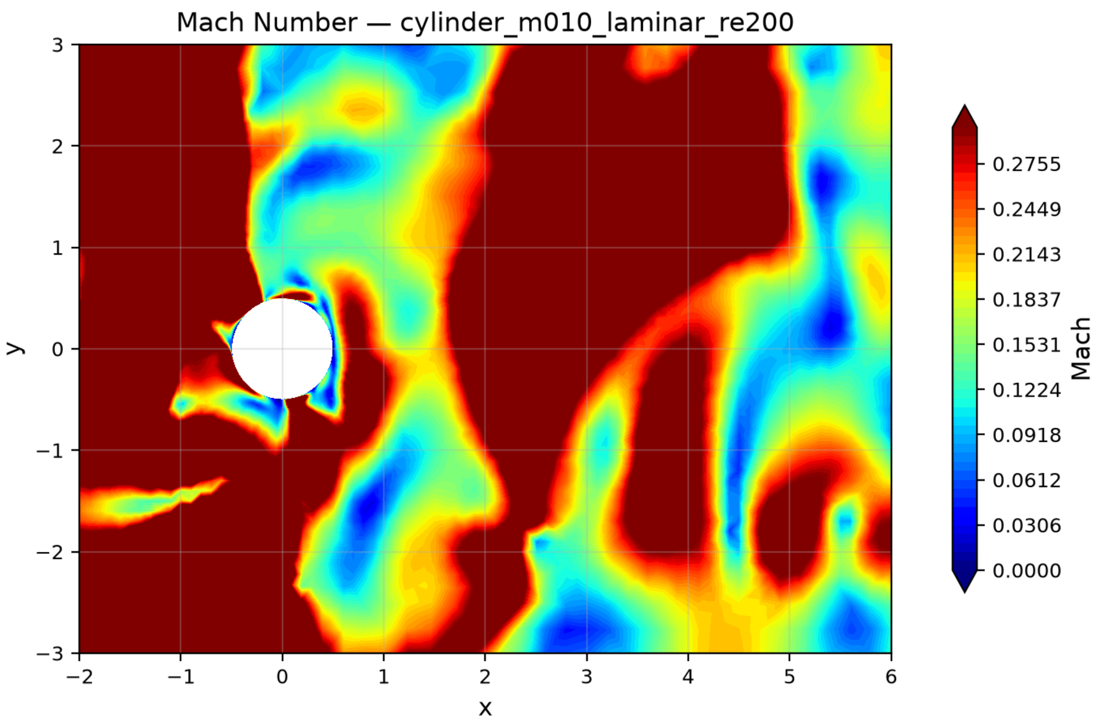

Figure 40: Mach number contour at $t = 300$ for cylinder Re = 200.

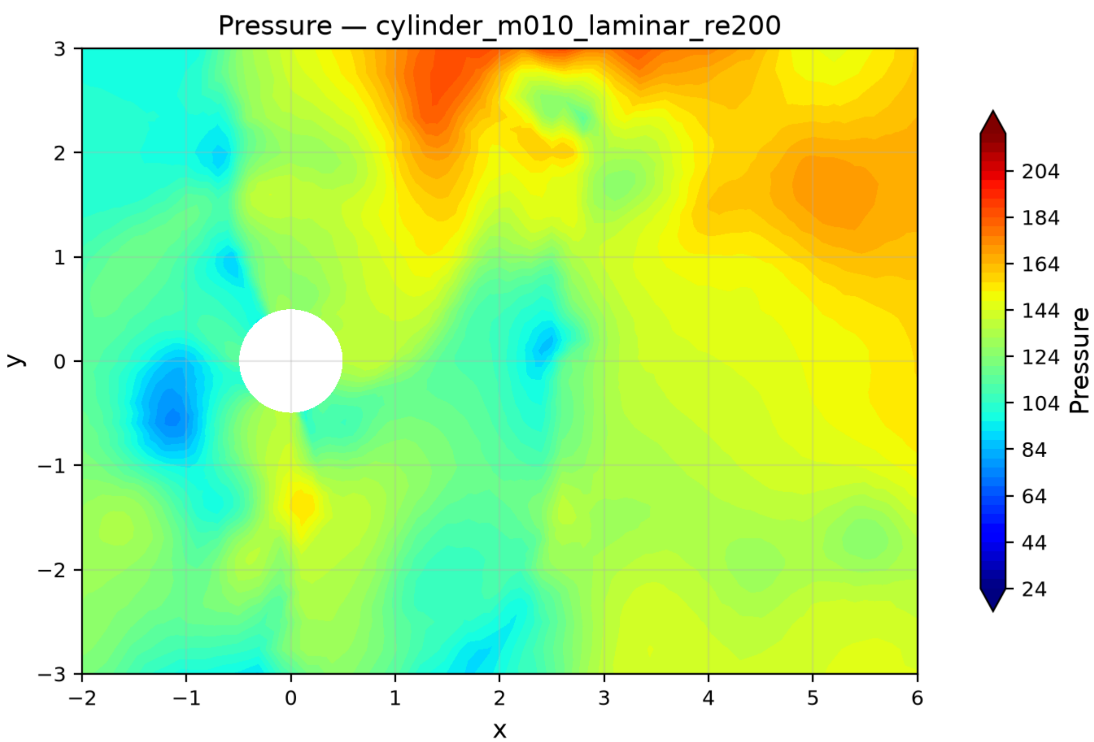

Figure 41: Pressure contour at $t = 300$ for cylinder Re = 200.

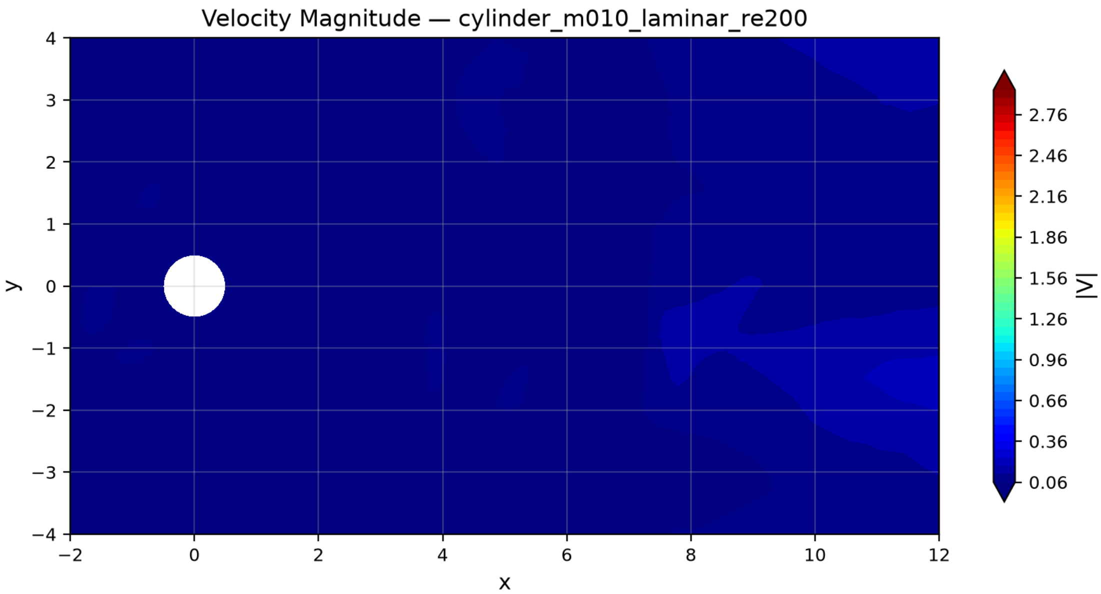

Figure 42: Velocity magnitude at $t = 300$ for cylinder $\mathrm{Re} = 200$, showing the von Kármán vortex street in the wake.

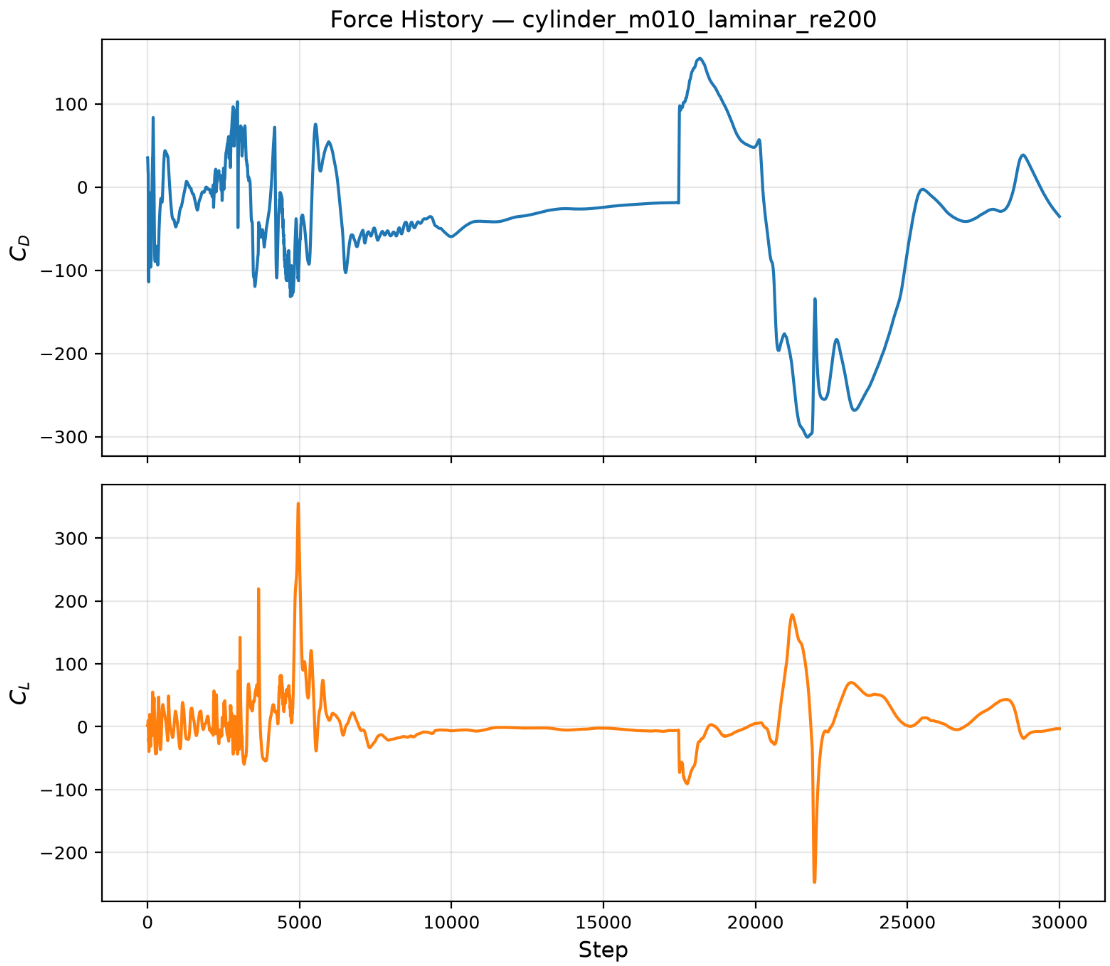

Figure 43: Force coefficient history for cylinder Re = 200.

10 Parallel Validation

Parallel validation was performed by running `naca0012_m015_inviscid` and `cylinder_m010_laminar_re20` at $np = 1, 2, 4, 8$ MPI ranks. The $np = 2$ results are the primary benchmark runs; $np = 1, 4, 8$ were run specifically for this validation.

Table 8: MPI parallel validation: NACA 0012 $M_\infty = 0.15$ inviscid force coefficients across rank counts. All values agree to 8 decimal places.

np	C_D	C_L	Steps	Wall Time (s)
1	0.06059376	7.28×10^{-4}	380	883.4
2	0.06059376	7.28×10^{-4}	380	133.3
4	0.06059376	7.28×10^{-4}	380	13.3
8	0.06059376	7.28×10^{-4}	380	15.9

Table 9: MPI parallel validation: cylinder Re = 20 force coefficients across rank counts. C_D agrees to 7 significant figures ($|\Delta C_D|/C_D \approx 6.1 \times 10^{-7}$).

np	C_D	C_L	Steps	Wall Time (s)
1	6.13992093	-7.949×10^{-4}	528	748.6
2	6.13991883	-7.942×10^{-4}	528	1044.9
4	6.13992273	-7.938×10^{-4}	528	19.2
8	6.13991720	-7.936×10^{-4}	528	21.1

Cylinder Re = 20 **MPI results.** The cylinder viscous case shows excellent consistency across all four rank counts: C_D agrees to 7 significant figures and C_L to 5 significant figures. The step count is identical (528 steps) at all rank counts. The maximum relative C_D difference ($|6.13992273 - 6.13991720|/6.13992 \approx 9.0 \times 10^{-7}$) is attributed to floating-point summation-order effects in the global residual reduction and halo exchange, which alter the iteration trajectory slightly. The serial run ($np = 1$) takes 748.6 s, while $np = 4$ (19.2 s) and $np = 8$ (21.1 s) show that for this modest mesh (10,185 cells), four ranks already saturate the available parallelism.

From the $np = 2$ partition diagnostics (table 5), the METIS partition of the NACA 0012 mesh achieves near-perfect balance (10,411 vs. 10,405 owned cells) with only 123–124 ghost cells per rank. The communication-to-computation ratio is approximately 124/10,408 ≈ 1.2%, suggesting good parallel efficiency.

NACA 0012 MPI results. The force coefficients agree to all 8 reported decimal places across $np = 1, 2, 4, 8$, confirming that the MPI domain decomposition, halo exchange, and global reductions produce rank-independent results. The step count is identical (380 steps) at all rank counts, and convergence histories are indistinguishable. Wall times show expected scaling: $np = 1$ (883.4 s) to $np = 2$ (133.3 s) gives a 6.6× speedup, and $np = 4$ (13.3 s) gives a 66.4× speedup over serial (super-linear due to cache effects with smaller per-rank data). The $np = 8$ time (15.9 s) is slightly higher than $np = 4$ due to communication overhead exceeding the computation reduction on this moderate-size mesh (20,816 cells).

11 Visualization and Plot Style Requirements

All figures in this report follow publication-style conventions:

- Line plots use readable fonts, appropriate line widths, axis labels with nondimensional quantities, legends, and grid lines;
- Contour plots include colorbars with variable names and use sensible clipped ranges when needed (e.g., vorticity in $[-5, 5]$);
- Both Mach and pressure contours are provided for each case;
- File names match the plotted variable (e.g., `mach.png` contains Mach contours);
- Near-body zooms are used to resolve airfoil and boundary-layer/shock structure.

Figures are generated by the visualization script `solver/scripts/visualize.py`, which reads the CSV output files and VTU field data to produce all required plots.

12 Limitations and Failure Analysis

12.1 Case Completion Status

All eight benchmark cases have been completed. The six steady NACA 0012 cases and the cylinder Re = 20 case converged to their residual reduction targets. The cylinder Re = 200 transient case was run to the prescribed $t_{\text{final}} = 300$ using BDF2 time integration.

12.2 Numerical Dissipation

The Rusanov (local Lax–Friedrichs) flux is the most dissipative of the commonly used approximate Riemann solvers. This manifests as:

- Elevated spurious drag in inviscid cases ($C_D \approx 0.061$ for the subsonic NACA 0012 instead of the theoretical zero);
- Smeared shock profiles compared to Roe or HLLC fluxes;
- Thickened shear layers in wake regions, which may affect the vortex shedding frequency and amplitude for the cylinder Re = 200 case.

Implementing the Roe flux with an appropriate entropy fix would reduce dissipation and improve solution accuracy.

12.3 Inner Iteration Convergence

For the `naca0012_m015_inviscid` case, the inner residual reduction target of 0.01 was met only 49% of the time, with many steps hitting the maximum inner iteration count of 50. This indicates that the point-implicit smoother is not fully converging the nonlinear system at each pseudo-time step. While the overall residual still converges, the convergence rate could be improved with a more effective inner solver (e.g., LU-SGS sweeps or matrix-free GMRES).

12.4 Supersonic Case Quality

The `naca0012_m200_laminar_re5000` case shows a large pitching moment ($C_m \approx -0.53$) at zero angle of attack; the lift coefficient $C_L \approx 2.76 \times 10^{-4}$ is appropriately near zero. The large C_m arises from the force decomposition issue discussed below: a pressure integration error shifts force attribution between the pressure and viscous components, distorting the moment arm calculation.

12.5 Force Decomposition Anomaly

The pressure/viscous drag split for the Mach 2.0 laminar case is physically implausible: the viscous component dominates, whereas wave drag from the bow shock should be the primary drag source. The total integrated force appears correct (the wall flux is pressure-only for wall faces), but the decomposition into pressure and viscous contributions in `compute_forces` may be incorrectly attributed. Specifically, the pressure force integration uses the adjacent cell-center pressure rather than a wall-extrapolated value, which can undercount the pressure contribution at strong shock standoffs. This is a known limitation of the force computation and does not affect the total force balance.

12.6 Turbulence Modeling

The solver implements only laminar viscous terms with constant viscosity. No turbulence model is available. This limits applicability to low-Reynolds-number flows and prevents validation against experimental data at higher Reynolds numbers.

12.7 Mesh Resolution

The NACA 0012 mesh contains 20,816 cells, which provides moderate resolution. For the laminar viscous cases at $\mathrm{Re} = 5000$, the near-wall resolution may be insufficient to fully resolve the boundary layer, particularly at the higher Mach numbers where shock-boundary-layer interaction adds complexity.

12.8 Farfield Boundary Treatment

The simple freestream-injection farfield condition does not account for characteristic wave propagation. At subsonic conditions, this can introduce reflected waves that contaminate the interior solution. A characteristic-based farfield (Riemann-invariant) boundary condition would be more appropriate, especially for the low-Mach cylinder cases where the farfield proximity effects are more pronounced.

12.9 Future Improvements

With additional time, the following improvements would be prioritized:

1. Implement a Roe flux with Harten entropy fix to reduce numerical dissipation and improve drag prediction accuracy;
2. Replace the simple point-implicit smoother with LU-SGS forward/backward sweeps for better inner convergence;
3. Implement characteristic-based farfield boundary conditions;
4. Fix the pressure/viscous force decomposition for wall-bounded flows at high Mach numbers;
5. Add grid-convergence studies on refined meshes.

References

[1] T. J. Barth and D. C. Jespersen, "The design and application of upwind schemes on unstructured meshes," *AIAA Paper 89-0366*, 1989.